Statements are tagged with the status of their proof: *Verified* means formalized in Lean 4 against Mathlib with no `sorry` and no axioms beyond `propext`, `Classical.choice` and `Quot.sound`; *Computed* means established by a finite computation, with the generating script cited. An untagged theorem, lemma, proposition or corollary carries a complete proof in the text and no formalization. Conjectures are labelled as such and are proved nowhere here. Definitions carry no status.

TWO- AND THREE-SPEED MODIFICATIONS OF THE TIGHT LONELY RUNNER INSTANCE: EXHAUSTIVE VERIFICATION, AND A LEAN 4 FORMALIZATION OF THE COVERING FRAMEWORK

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Abstract. For a set V of $n-1$ distinct positive integers write $\mathrm{LR}(V) = \max_t \min_{v\in V} \|vt\|$, where $\|x\|$ is the distance from x to the nearest integer; V is *tight* if $\mathrm{LR}(V) = 1/n$, the value predicted by the Lonely Runner Conjecture. Every such V is a *k-speed modification* of the baseline $[n-1] = \{1,\dots,n-1\}$: writing $M = [n-1]\setminus V$ and $R = V \setminus [n-1]$ one has $|M| = |R| = k$, where k counts the speeds of V that are at least n.

The case $k = 1$ is settled. Goddyn and Wong [6] classified the tight sets obtained by replacing one speed with a multiple, and Zhang [17] (version 2, 31 August 2026) completed the classification of one-entry modifications (Thm. 1.2), bounded the largest speed by $O(n\log n)$ (Thm. 1.3), and computed exactly the components of the region $U(n,r)$ left uncovered when a single speed r is deleted (Thm. 1.4). For $k = 2$, Zhang [17, Thms. 7.5, 7.10] proves that no tight two-speed modification exists for removed sets $\{2,r'\}$ with $n \geq 19$ and $\{q,r'\}$ with $n \geq \max(40, 10q)$, with no bound on the inserted speeds, and his census [17, Thm. 7.13] covers $8 \leq n \leq 140$ with inserted speeds at most $8n$. Version 1 of that paper had verified $6 \leq n \leq 34$ under the hypothesis that the inserted speeds are at most $4n$.

This paper contributes three things. First, a covering bound which is uniform over the inserted speeds: if $\mathrm{LR}(V) \leq 1/n$ then $\sum_{v\in R} 1/v \;\geq\; \ell(n,M)(n-2k)/2$, where $\ell(n,M)$ is the length of the longest component of the uncovered region, with *no* upper bound on any inserted speed assumed. For $k = 2$ Zhang's non-existence theorems [17, Thms. 7.5, 7.10] are stronger than the bound here on the removed sets they cover; the measure argument here is uniform in k and in the inserted speeds, at the cost of a constant. Second, and as a consequence, an exhaustive verification with the cap on the inserted speeds removed: the conjecture holds for every 2-speed modification with $5 \leq n \leq 100$ and every 3-speed modification with $7 \leq n \leq 40$. The enumeration is finite for two rigorous reasons and no branch was truncated. Third, a formalization in Lean 4 with Mathlib of the covering framework and of the description of the uncovered region, comprising 70 theorems and 5 definitions with no `sorry`. We are not aware of any previous formalization of the material treated here; for the conjecture itself, the Acorn library contains proofs of statements with the constant slack by one, discussed in the companion paper. The statement itself, and Tao's bound, appear in Lean in the `formal-conjectures` repository with `sorry` bodies and no proof content. For the wider context of the problem we refer to the survey [10]. Not every statement below is formalized, and the Formalization section is the authoritative account of which are and which are not; downstream of the strip lemma the formalized constants carry $2 + 4/n$ in place of the sharp 2.

Much of Sections 3, 4, 5 and 8 is expository; Section 7 and the formalization are not. The description of the uncovered region by localization at fractions whose denominators were deleted is Zhang's [17, §2, proof of Lemma 2.4], used constructively by Bohman and Peng [2, §3]; the measure of the uncovered region as an object of study is Cusick's [4], with measure and component-count bookkeeping already in Cusick and Pomerance [5]; the gap-and-block framework is Bohman, Holzman and Kleitman [1]; the divisibility obstruction is [6, Lemma 2.2], [17, Lemmas 2.3, 6.1] and [12, Lemma 4]; and the factor-of-two barrier for measure arguments is [15, Prop. 1.1]. We include these because the formalization and the search consume them, and we claim no priority for any of them.

1. Introduction

Let $\|x\|$ be the distance from $x \in \mathbb{R}$ to the nearest integer. For a set V of $n-1$ distinct positive integers put $\mathrm{LR}(V) = \max_{t\in\mathbb{R}} \min_{v\in V} \|vt\|$. The Lonely Runner Conjecture asserts $\mathrm{LR}(V) \geq 1/n$. The baseline $[n-1] = \{1,\dots,n-1\}$ satisfies $\mathrm{LR}([n-1]) = 1/n$, attained at $t = 1/n$, so it is *tight*.

Goddyn and Wong [6] classified the tight sets obtained from $[n-1]$ by replacing one speed r with a multiple, and proved that for fixed r only finitely many non-multiple replacements can be tight. Zhang [17] made that finiteness effective. Writing $U(n,r)$ for the region left uncovered when r is deleted (Definition 1.3 below, with $M = \{r\}$), he computes the length of every connected component of $U(n,r)$ exactly, in both regimes $2r > n-1$ and $2r \leq n-1$, and

deduces an explicit upper bound on the inserted speed, hence $n \le 6r$ when $2r \le n-1$, together with the complete classification for $r = 2$ and $r = 3$.

Throughout, $P(b)$ denotes the smallest prime not dividing b.

This note treats modifications of arbitrary size.

Definition 1.1. Let $n \ge 3$ and $k \ge 1$. A *k-speed modification* of the baseline is

$$V \;=\; \big([n-1] \setminus M\big) \cup R, \qquad M \subseteq [n-1],\ |M| = |R| = k,\ \ v \ge n \text{ for all } v \in R.$$

Remark 1.2. This is no restriction on the speed set. Let V be any set of $n-1$ distinct positive integers, and put $M = [n-1] \setminus V$, $R = V \setminus [n-1]$, $k = \#\{v \in V : v \ge n\}$. Then $|V \cap [n-1]| = n-1-k$, so $|M| = (n-1) - (n-1-k) = k = |R|$, and V is the k-speed modification determined by M and R. Every instance with $1 \le k \le n-2$ arises exactly once in this way, with k recording how many speeds exceed the baseline. The hypothesis $2k < n$ belongs to Theorem 1.4, not to the setting: Theorem 1.10 holds for every $k \le n-2$, and Theorem 5.2 for every $k \le n-2$ as well.

Then $|V| = n-1$, so the predicted floor for V is again $1/n$. The inserted speeds are subject to no upper bound, and that is the difficulty.

Definition 1.3. For $M \subseteq [n-1]$ and $v \ge 1$ put

$$U(n,M) = \big\{t \in [0,1) : \|jt\| \ge 1/n \ \text{ for all } j \in [n-1] \setminus M\big\}, \qquad B_v = \big\{t \in [0,1) : \|vt\| < 1/n\big\}.$$

Both depend on n; the dependence of B_v on n is suppressed. Since $U(n,M)$ is a finite union of closed intervals, a longest component exists; write $\ell(n,M)$ for its length, which is 0 exactly when $U(n,M)$ is finite.

$U(n,M)$ is closed and each B_v is open of Lebesgue measure exactly $2/n$, independently of v. Directly from the definitions, $\mathrm{LR}(V) \ge 1/n$ if and only if $U(n,M) \not\subseteq \bigcup_{v\in R} B_v$.

What is new here, and what is not. Arguments that show a family of bad sets fails to cover the uncovered region of a sub-family are long established. Bohman, Holzman and Kleitman [1] work with exactly this dichotomy, calling a maximal uncovered interval for a sub-collection S an S-gap and a maximal covered interval an S-block; their Lemma 4 bounds the length of a block of the *two* fastest speeds and their Corollary 5 concludes non-covering from a long enough gap. So the passage from one covering speed to two is not new, and neither is the idea of measuring the uncovered region, which goes back to Cusick [4] and, with component counting alongside the measure, to Cusick and Pomerance [5].

What is new is the form of the bound. For a single inserted speed the natural step is a connectivity argument: a connected subset of $\{t : \|wt\| \le 1/n\}$ lies inside a single one of its constituent intervals, so its length is at most $2/(wn)$. That uses in an essential way that there is only *one* inserted speed. The argument of [1] handles two, but through explicit block-length estimates tied to the five- and six-runner case analysis. Section 7 of [17] (version 2) handles two inserted speeds by a counting bound on the part of a bad set inside a component (Lemma 7.3) and proves the non-existence statements of Theorems 7.5 and 7.10 with no bound on the inserted speeds; on the removed sets they cover, for $k = 2$, these are stronger than Theorem 1.4, which only bounds $\sum_{v\in R} 1/v$. Theorem 1.4 replaces these arguments by a measure estimate which holds for every k simultaneously, is uniform over the inserted speeds, and assumes no upper bound on any of them. That uniformity is what makes the enumeration of Section 7 finite for $k = 3$ without a cap on $\max R$; for $k = 2$ the cap that version 1 of [17] needed is removed there by other means.

The price is a constant. Theorem 1.4 carries a factor 2 that the connectivity argument does not, so at $k = 1$ it is weaker than [17, Thm. 1.4], which computes $\ell(n,\{b\})$ exactly. Sections 3

to 4 accordingly record the $k = 1$ theory in the form the later sections consume, with attribution; the reader wanting the sharp $k = 1$ statements should take them from [17].

Theorem 1.4 (Proved; Covering bound)**.** *Let V be a k-speed modification with $2k < n$ and $\mathrm{LR}(V) \le 1/n$. Then*

$$\sum_{v\in R} \frac{1}{v} \;\ge\; \frac{\ell(n,M)\,(n-2k)}{2}, \qquad \textit{and consequently} \qquad \min R \;\le\; \frac{2k}{\ell(n,M)\,(n-2k)}$$

whenever $\ell(n, M) > 0$. No upper bound on any element of R is assumed.

A large inserted speed is close to equidistributed and contributes about $2\ell/n$ to a component of length ℓ; k of them contribute about $2k\ell/n < \ell$ and cannot finish the covering. The deficit must come from the boundary term $2/(vn)$, which is large only for small v.

The remaining task is to bound $\ell(n, M)$ below, uniformly in M. We do this from scratch in Section 4, by exhibiting an explicit interval inside the uncovered region; the following observation reduces every k to the case of a single deleted speed.

Lemma 1.5 (Proved; Monotonicity)**.** *If $M' \subseteq M$ then $U(n, M') \subseteq U(n, M)$; hence for nonempty M, $\ell(n, M) \ge \max_{b\in M} \ell(n, \{b\})$.*

Theorem 1.6 (Proved; Uniform component bound)**.** *Let $n \ge 3$ and let $M \subseteq [n-1]$ be nonempty.*

(i) *$\ell(n, M) \;\ge\; \dfrac{1}{2n^3}$, with an elementary and explicit proof.*

(ii) *If M contains an element coprime to n, or an element b with $2b > n-1$, then $\ell(n, M) \;\ge\; \dfrac{1}{n^2\,(2\log n + 2)}$, again explicitly.*

(iii) *$\ell(n, M) \;\ge\; c/(n^2 \log n)$ for an absolute constant c. Here c is not explicit: the remaining case rests on Iwaniec's bound for Jacobsthal's function, whose implied constant is effective but has not been computed.*

In the coprime case the truth is sharper, and the sharp constant is attained.

Theorem 1.7 (Proved; Coprime component bound)**.** *Let $n \ge 2$, let $1 \le a < n$ be coprime to n, and let $j_+ \in [1, n)$ be the unique speed with $j_+ a \equiv 1 \pmod n$. If $j_+ \in M$ then*

$$\left[\tfrac{a}{n} - \tfrac{1}{n(n-1)},\ \tfrac{a}{n}\right] \;\subseteq\; U(n, M), \qquad \textit{so} \qquad \ell(n, M) \;\ge\; \frac{1}{n(n-1)}.$$

In particular this holds whenever M contains an element coprime to n, and for every nonempty M when n is prime. The constant cannot be improved: Proposition 6.1 exhibits equality, computationally.

Combining Theorems 1.4, 1.6 and 1.7:

Corollary 1.8 (Proved)**.** *Let V be a k-speed modification with $2k < n$ and $\mathrm{LR}(V) \le 1/n$. Then*

$$\min R \;\le\; \frac{4k\,n^3}{n-2k},$$

and if M contains an element coprime to n,

$$\min R \;\le\; \frac{2k\,n(n-1)}{n-2k}.$$

For n prime the second bound is unconditional in M. For $k = 1$ we have $\min R = \max R$, so these bound the inserted speed outright.

Proof. Immediate from Theorems 1.4, 1.6 and 1.7: the first supplies the covering inequality, the second the component bound valid for every M, and the third the sharper bound when some element of M is coprime to n. For n prime every element of M not divisible by n is coprime to it, and $M \subseteq [n-1]$, so the coprime case always applies. □

Remark 1.9. We state inequalities rather than asymptotics deliberately. Definition 1.1 permits $n - 2k = 1$, and there the second bound of Corollary 1.8 is $\Theta(n^3)$, not $\Theta(kn)$, so an assertion of the form $\min R = O(kn)$ is not available from these bounds. We do not claim it is false: we know of no family realising $\min R \gg kn$, and a search over tight modifications with $n - 2k = 1$ found $\max \min R = 6$ at $n = 5$, $k = 2$ and 8 at $n = 7$, $k = 3$. That search is finite for a reason internal to this paper: Theorem 1.4 bounds the smallest inserted speed, and applying it again to the region left after that speed is removed bounds the next, a branch terminating as soon as the region has no component of positive length. At such a leaf the branch is scored without further search, provided the region is nonempty, which holds in every case we searched and which for fewer than $n-1$ speeds is an instance of the conjecture itself. The region is then a finite set of rationals; any $t = p/q$ in it, in lowest terms, has $q \geq 2$ because $0 \in B_j$ for every present speed j and $[n-1] \setminus M \neq \emptyset$. One may take every remaining speed in the class $v \equiv c \pmod q$ with $cp \equiv \lfloor q/2 \rfloor$, giving $\|vt\| = \lfloor q/2 \rfloor / q \geq 1/3$; infinitely many such v exist, so t survives and $\min R$ is already determined. Under the additional hypothesis $2k \leq (1-\varepsilon)n$ one does get $\min R = O_\varepsilon(kn)$ in the coprime case, $O_\varepsilon(kn^2)$ in general, and $O_\varepsilon(kn \log n)$ from Theorem 1.6(iii).

Bounding $\ell(n, M)$ from below by exhibiting an interval, as above, leaves open how much of the uncovered region those intervals miss. They miss nothing: the region has no freedom at all in where it can sit. The reason is Dirichlet's approximation theorem, which for every real t produces a denominator $q \leq n-1$ approximating t to within $1/(qn)$. If t is uncovered then that denominator cannot be a present speed, so it was deleted.

Theorem 1.10 (Proved; Localization)**.** *Let $n \geq 2$ and $M \subseteq [n-1]$, and let $t \in \mathbb{R}$ satisfy $\|jt\| > 1/n$ for every present j. Then there are $q \in M$ and $a \in \mathbb{Z}$ with $\gcd(a, q) = 1$ and*

$$\left| t - \frac{a}{q} \right| \;\leq\; \frac{1}{qn}.$$

Since there are $\sum_{q \in M} q$ fractions with denominator in M, each carrying an interval of length $2/(qn)$, this bounds the whole region at once, uniformly in how large the deleted speeds are.

Corollary 1.11 (Proved)**.** $\lambda\big(U(n, M)\big) \;\leq\; 2|M|/n$.

More is true. Each component of $U(n, M)$ not only lies near such a fraction a/q but runs back to it, so the component is cut out by an explicit minimum over the present speeds. That is Theorem 5.2: writing $w(n, b, a, M)$ for the window at a/b of Definition 5.1, and provided $M \neq [n-1]$ so that some speed is present,

$$\ell(n, M) \;=\; \max_{b \in M} \; \max_{\substack{0 \leq a < b \\ \gcd(a,b)=1}} \; w(n, b, a, M),$$

an identity costing $O\big(n \sum_{b \in M} \varphi(b)\big)$ operations rather than an $O(n^2 \log n)$ sweep. Feeding it into Theorem 3.1 gives Corollary 5.4, an unconditional criterion which, unlike Corollary 3.2, loses nothing: over all k-subsets at $n = 22, \ldots, 42$ and $k = 1, 2, 3$ it certifies exactly the subsets for which the criterion of Theorem 3.1 holds at all.

The coprimality hypothesis in Theorem 1.7 cannot be dropped, and the witness is the Goddyn–Wong set recorded in version 1 of [17] (it is not in version 2): $\{1, \ldots, 29, 31, 90\}$ has $n = 32$, $M = \{30\}$, $R = \{90\}$ and $\gcd(30, 32) = 2$. Here $\ell(32, \{30\}) = 1/2576$, the bound of Theorem 1.4 reads $\min R \leq 171.73$ and holds, while the coprime bound of Corollary 1.8 reads $\min R \leq 66.13$ and fails.

2. The covering bound

Lemma 2.1 (Proved)**.** *Let C be an interval of length ℓ, let $n \geq 2$ and let $v \geq 1$ be an integer. Then $\lambda(B_v \cap C) \leq 2\ell/n + 2/(vn)$, where λ is Lebesgue measure.*

Proof. Work with the 1-periodic extension of B_v to $\mathbb{R}$, which is legitimate since intersecting back with $[0,1)$ only decreases measure. It is invariant under translation by $1/v$, and each period carries measure exactly $2/(vn)$: the map $t \mapsto vt$ sends a window of length $1/v$ bijectively onto a window of length 1, and $\lambda\{u \in [c, c+1) : \|u\| < 1/n\} = 2/n$. Put $m = \lceil \ell v \rceil$ and $C = [x, x+\ell]$. Then $C \subseteq [x, x + m/v]$, a union of m windows of length $1/v$, so $\lambda(B_v \cap C) \leq m \cdot 2/(vn) \leq (\ell v + 1) \cdot 2/(vn) = 2\ell/n + 2/(vn)$. □

Proof of Theorem 1.4. First, $\mathrm{LR}(V) \leq 1/n$ forces $S = \{t : \|vt\| \geq 1/n \text{ for all } v \in V\}$ to be a null set: for $t \in S$ the minimum $\min_{v \in V} \|vt\|$ is at least $1/n$ and, by hypothesis, at most $1/n$, so it equals $1/n$ and some $v \in V$ has $\|vt\| = 1/n$. Hence $S \cap [0,1) \subseteq \bigcup_{v \in V} \{t \in [0,1) : \|vt\| = 1/n\}$, a finite set. So $U(n, M)$ is covered by $\bigcup_{v \in R} B_v$ up to measure zero.

Let $C \subseteq U(n, M)$ be a component of length $\ell = \ell(n, M) > 0$. Then $\ell \leq \sum_{v \in R} \lambda(B_v \cap C)$, so Lemma 2.1 gives

$$\ell \;\leq\; \sum_{v \in R} \Big(\frac{2\ell}{n} + \frac{2}{vn} \Big) \;=\; \frac{2k\ell}{n} + \frac{2}{n} \sum_{v \in R} \frac{1}{v},$$

whence $\ell(n - 2k) \leq 2 \sum_{v \in R} 1/v$. Since $|R| = k$, $\sum_{v \in R} 1/v \leq k / \min R$. □

Remark 2.2. The hypothesis $v \geq n$ of Definition 1.1 is not used in this proof; Lemma 2.1 needs only $v \geq 1$. It is needed for $|V| = n - 1$, not for the argument.

3. An unconditional case

Theorem 1.4 says that covering forces $\sum_{v \in R} 1/v$ to be large. Since every inserted speed is at least n, that sum is at most k/n. When the two are incompatible the covering hypothesis is untenable and no R whatsoever can cover, so V is lonely with no bound on any inserted speed and no appeal to anything further.

Theorem 3.1 (Proved; Unconditional loneliness)**.** *Let $n \geq 3$ and $k \geq 1$ with $2k < n$, and let $M \subseteq [n-1]$ with $|M| = k$. If*

$$\ell(n, M) \;>\; \frac{2k}{n\,(n - 2k)},$$

then $\mathrm{LR}\big(([n-1] \setminus M) \cup R\big) \geq 1/n$ for every *admissible R, that is, for every R as in Definition 1.1.*

Proof. If $\mathrm{LR}(V) < 1/n$ then Theorem 1.4 gives $\sum_{v \in R} 1/v \geq \ell(n, M)(n - 2k)/2 > k/n$. But $|R| = k$ and every $v \in R$ satisfies $v \geq n$, so $\sum_{v \in R} 1/v \leq k/n$, a contradiction, since Theorem 1.4 applies as soon as $\mathrm{LR}(V) \leq 1/n$. Hence $\mathrm{LR}(V) \geq 1/n$. □

Combined with the lower bounds of Section 4 this settles an explicit subclass outright. Feeding Proposition 4.1's bound $\ell(n, \{b\}) \geq (2s - j_0)/(2nbj_0)$, valid for $2b \leq n-1$, and Proposition 4.2's bound $\ell(n, \{b\}) \geq \frac{s}{nb}(\frac{1}{j_0} + \frac{1}{n-1})$, valid for $2b \geq n$, into the criterion, and using Lemma 1.5:

Corollary 3.2 (Proved)**.** *Let $n \geq 3$ and $k \geq 1$ with $2k < n$, and let $M \subseteq [n-1]$ with $|M| = k$. Write $s = n - b$, and for $b \geq 2$ let $j_0 = j_0(n, b)$ be as in Section 4 (case (a) below does not use it). Suppose M contains a speed b with either*

(a) *$b = 1$ and $n > 6k$; or*
(b) *$2 \leq b$, $2b \leq n - 1$ and $(2s - j_0)(n - 2k) > 4kb\,j_0$; or*
(c) *$2 \leq b$, $2b > n - 1$ and $s(n - 2k)(j_0 + n - 1) > 2kb\,j_0(n - 1)$.*

Then $\mathrm{LR}(([n-1] \setminus M) \cup R) \geq 1/n$ for every admissible R.

Proof. Case (a) is separate because Proposition 4.1 requires $b \geq 2$: the integer a there lies in $[1, b)$, which is empty at $b = 1$. It is handled directly, and the argument is that of [6, proof of Thm. 2.3, case $r = 1$]. Every present speed j satisfies $2 \leq j \leq n-1$, so for $t \in [\frac{1}{2n}, \frac{1}{n}]$ we have $jt \in [\frac{j}{2n}, \frac{j}{n}] \subseteq [\frac{1}{n}, 1 - \frac{1}{n}]$, whence $\|jt\| \geq 1/n$. Thus $[\frac{1}{2n}, \frac{1}{n}] \subseteq U(n, \{1\})$ and $\ell(n, \{1\}) \geq \frac{1}{2n}$, which exceeds $2k/(n(n-2k))$ exactly when $n > 6k$. (In fact $\ell(n, \{1\}) = 1/(2n)$; we verified this for $3 \leq n \leq 44$.)

In case (b), Proposition 4.1 gives $\ell(n, \{b\}) \geq (2s - j_0)/(2nbj_0)$; in case (c), Proposition 4.2 gives $\ell(n, \{b\}) \geq \frac{s}{nb}(\frac{1}{j_0} + \frac{1}{n-1})$. The stated inequalities say exactly that these exceed $2k/(n(n-2k))$. Lemma 1.5 raises the bound from $\ell(n, \{b\})$ to $\ell(n, M)$, and Theorem 3.1 applies. □

Remark 3.3. Condition (b) admits a closed form at the cost of some strength. Since $j_0 \leq n-1$ one has $(2s-j_0)/j_0 \geq (n-2b+1)/(n-1)$, so the inequality clause of (b) follows from $4kb(n-1) < (n-2b+1)(n-2k)$, and that in turn from $8kb \leq n$ together with $n > 6k-2$. The side clauses of (b) come along too when $b \geq 2$: $8kb \leq n$ gives $2b \leq n/(4k) \leq n/4 \leq n-1$. (At $b = 1$ the hypotheses land in case (a) instead.) The derivation: the first gives $4kb(n-1) \leq n(n-1)/2$, while $2b \leq n/(4k) \leq n/4$ gives $(n-2b+1)(n-2k) \geq \frac{3}{4}n(n-2k)$, and $n(n-1)/2 < \frac{3}{4}n(n-2k)$ reduces to $n > 6k-2$. We verified $4kb(n-1) < (n-2b+1)(n-2k)$ directly on all 4 338 029 triples with $b \geq 2$, $n \leq 4000$ and $k \leq 60$ admitted by those hypotheses, with no exception. (Admitting $b = 1$, which the previous sentence sends to case (a) instead, the count is 4 563 449 and there is still no exception.)

The step $j_0 \leq n-1$ is wasteful for mid-range b, where j_0 exceeds s only by the gap to the next integer coprime to b. If that gap is $o(s)$ then (b) reads $n - 2k > 4kb$ asymptotically, i.e. $b < (n-2k)/(4k)$, roughly twice the range of the closed form. We do not use this, since bounding the gap needs the input of Theorem 1.6(iii); conditions (b) and (c) are exact and checkable as they stand.

How much this settles, counted exactly over *all* k-subsets M. Over $22 \leq n \leq 38$ the criterion of Theorem 3.1 holds for 36.00% to 44.44% of subsets when $k = 1$, for 17.66% to 25.08% when $k = 2$, and for 11.88% to 19.56% when $k = 3$. Its reach falls away for small n: at $n = 10$ it settles 22.22% of subsets when $k = 1$ and nothing at all when $k = 2$ or $k = 3$, the first n at which it is satisfiable being $7, 11, 15$ for $k = 1, 2, 3$. Corollary 3.2 recovers most of that. At $n = 22, 26, 30, 34, 38, 42$, with the criterion's own reach printed alongside so that the comparison is on the same n:

n	22	26	30	34	38	42
criterion, $k = 1$	38.10	36.00	41.38	42.42	37.84	41.46
Cor. 3.2, $k = 1$	33.33	32.00	37.93	33.33	35.14	36.59
ratio	0.875	0.889	0.917	0.786	0.929	0.882
criterion, $k = 2$	21.43	19.33	23.89	21.78	24.77	22.44
Cor. 3.2, $k = 2$	18.57	15.67	19.95	17.61	20.72	18.78
ratio	0.867	0.811	0.835	0.808	0.836	0.837
criterion, $k = 3$	14.44	13.13	11.88	18.37	16.82	15.55
Cor. 3.2, $k = 3$	14.29	12.00	10.34	17.61	15.77	14.27
ratio	0.990	0.914	0.870	0.959	0.938	0.918

The closed form of Remark 3.3 is weaker, certifying $9.52, 12.00, 10.34, 12.12, 10.81, 12.20$ per cent at $k = 1$ over the same n. The residual gap is closed outright in Section 5: Corollary 5.4 replaces the case analysis by a single explicit window and matches the criterion in every cell of the table above.

The two-sided window of Proposition 4.2 is what makes case (c) useful at $k = 1$. With the one-sided bound of Proposition 4.1 in its place, case (c) is provably empty at every k: in that regime

$2b \geq n$ and $j_0 \geq s$ give $2kbj_0 \geq kns \geq ns > (n-2k)s$, so the one-sided inequality $s(n-2k) > 2kbj_0$ has no solutions at all, and the $k=1$ reach falls to $19.05, 20.00, 20.69, 21.21, 21.62, 21.95$ per cent. Even with the two-sided window, case (c) remains empty for every $k \geq 2$; see Remark 5.5.

The reason the criterion fires so often is that $\ell(n, \{b\})$ is of order $1/(2bn)$ for small b, not of order $1/n^2$: at $n = 40$ the values of $\ell\, n^2$ for $b = 1, \ldots, 9$ are $20.000, 9.487, 6.667, 4.730, 3.778, 3.143, 2.857, 2.348, 2.222$, in close agreement with $n/(2b)$. This is what Proposition 4.1 predicts, since $s \approx n$ and $j_0 \approx n$ there.

4. The uncovered region for general M

Proof of Lemma 1.5. $[n-1] \setminus M \subseteq [n-1] \setminus M'$, so the constraints defining $U(n, M)$ are a subfamily of those defining $U(n, M')$. Taking $M' = \{b\}$ for $b \in M$ gives the second claim. □

By Lemma 1.5 it suffices to exhibit a long interval inside $U(n, \{b\})$ for a single b. We do this directly. Fix $2 \leq b < n$, put $s = n - b \geq 1$ and

$$j_0 \;=\; j_0(n, b) \;=\; \min\{\, j \geq s :\ \gcd(j, b) = 1 \,\}.$$

The window $[s, s+b-1]$ consists of b consecutive integers, one of them congruent to 1 modulo b, so j_0 exists and $s \leq j_0 \leq s+b-1 = n-1$. In particular j_0 is a speed of $[n-1]$, and $j_0 \neq b$ because $\gcd(j_0, b) = 1$ and $b \geq 2$.

Proposition 4.1 (Proved; Missing-speed interval)**.** *Let $2 \leq b < n$ and let a be the unique integer with $1 \leq a < b$ and $j_0\, a \equiv -1 \pmod b$; it exists because $\gcd(j_0, b) = 1$, and $a \neq 0$ because $\gcd(a, b) = 1$. Put*

$$\delta_0 = \begin{cases} \dfrac{1}{2bn}, & 2b \leq n-1, \\ 0, & 2b > n-1. \end{cases}$$

Then the interval below is nonempty: if $2b \leq n-1$ then $s \geq b+1$, so $j_0 \leq s+b-1 \leq 2s-2$ and $\delta_0 < s/(nbj_0)$; while if $2b > n-1$ then $\delta_0 = 0$. (The claim $j_0 \leq 2s$ is false in the second regime: at $n = 32$, $b = 30$ one has $s = 2$ and $j_0 = 7$.) Moreover $\left[\frac{a}{b} + \delta_0,\ \frac{a}{b} + \frac{s}{nbj_0}\right] \subseteq U(n, \{b\}) \subseteq [0, 1)$; hence

$$\ell(n, \{b\}) \;\geq\; \frac{s}{n\, b\, j_0} - \delta_0 \;=\; \begin{cases} \dfrac{2s - j_0}{2\, n\, b\, j_0}, & 2b \leq n-1, \\ \dfrac{s}{n\, b\, j_0}, & 2b > n-1. \end{cases}$$

Proof. Let $\delta \in [\delta_0,\ s/(nbj_0)]$ and $t = a/b + \delta$. Note $\delta \leq 1/(nb)$, since $j_0 \geq s$. Let $j \in [1, n)$ with $j \neq b$.

First, $t \in [0, 1)$: since $1 \leq a < b$ we have $t \geq a/b \geq 1/b > 0$, and $t \leq \frac{b-1}{b} + \frac{1}{nb} = 1 - \frac{n-1}{nb} < 1$.

Multiples of b. If $b \mid j$, write $j = mb$; then $m \neq 0$ and $m \neq 1$, so $m \geq 2$ and $j \geq 2b$, which forces $2b \leq n-1$ and hence $\delta_0 = 1/(2bn)$. Now $j \cdot (a/b) = ma \in \mathbb{Z}$, so $\|jt\| = \|j\delta\|$, and

$$j\delta \;\geq\; 2b \cdot \frac{1}{2bn} = \frac{1}{n}, \qquad j\delta \;\leq\; (n-1) \cdot \frac{1}{nb} \;\leq\; \frac{n-1}{n} = 1 - \frac{1}{n}.$$

A real number in $[1/n, 1 - 1/n]$ is at distance at least $1/n$ from every integer.

Non-multiples. If $b \nmid j$, put $r = ja \bmod b$; then $r \neq 0$ because $\gcd(a, b) = 1$, so $1 \leq r \leq b-1$. Writing $ja = bQ + r$ gives $jt = Q + \left(\frac{r}{b} + j\delta\right)$, and $\frac{r}{b} + j\delta \geq \frac{1}{b} \geq \frac{1}{n}$. For the upper edge we must check $j\delta \leq \frac{b-r}{b} - \frac{1}{n}$, and it suffices to check it at $\delta = s/(nbj_0)$. If $r = b-1$ then $j \equiv -a^{-1} \equiv j_0 \pmod b$; the elements of that class below n are $j_0,\ j_0 - b,\ j_0 - 2b, \ldots$, since $j_0 + b \geq s + b = n$, so $j \leq j_0$ and

$$j\delta \;\leq\; j_0 \cdot \frac{s}{nbj_0} = \frac{s}{nb} = \frac{1}{b} - \frac{1}{n} = \frac{b-r}{b} - \frac{1}{n}.$$

If $r \le b-2$ then, using $j \le n-1$ and $s \le j_0$,

$$j\delta \;\le\; \frac{(n-1)s}{nbj_0} \;\le\; \frac{n-1}{nb} \;\le\; \frac{2n-b}{nb} \;=\; \frac{2}{b}-\frac{1}{n} \;\le\; \frac{b-r}{b}-\frac{1}{n},$$

the middle inequality because $b \le n+1$. So no residue class with $b-r \ge 2$ is ever binding. □

Proposition 4.2 (Proved; Two-sided window)**.** *Let $2 \le b < n$ with $2b > n-1$, and let a, j_0 be as in Proposition 4.1. Then no multiple of b other than b itself is a speed of $[n-1]$, the window extends on both sides of a/b, and*

$$\ell(n,\{b\}) \;\ge\; \frac{s}{n\,b}\Big(\frac{1}{j_0}+\frac{1}{n-1}\Big).$$

Proof. Since $2b > n-1$, any multiple mb with $m \ge 2$ satisfies $mb \ge 2b \ge n$, so the only present speeds are non-multiples of b, and Proposition 4.1 applies with $\delta_0 = 0$ on the right. On the left, take $t = a/b-\delta$ with $\delta \ge 0$. For a present j with $r = ja \bmod b$, the argument of Proposition 4.1 applies with the roles of r and $b-r$ exchanged: the binding class is now $r=1$, that is $j \equiv a^{-1} \pmod b$, and the binding member is the largest such j below n, which is at most $n-1$. Hence $\delta \le s/(nb(n-1))$ is admissible, and the interval $[\,a/b - s/(nb(n-1)),\ a/b + s/(nbj_0)\,]$ lies in $U(n,\{b\})$. □

We verified this bound against the exhaustively computed $\ell(n,\{b\})$ for all 9999 pairs with $2 \le b < n \le 200$ and $2b \ge n$: no violation. Equality holds throughout the family $b = n/2$, n even, which we verified for all even $6 \le n \le 250$; over all admissible pairs with $n \le N$ the proportion of exact cases falls from 14.3% at $N = 30$ to 1.8% at $N = 250$. The bound is exact, for instance $\ell(8,\{4\}) = 3/70 = 1/40 + 1/56$ and $\ell(32,\{16\}) = 1/544 + 1/992$.

Remark 4.3. Proposition 4.2 lives in the regime $2b \ge n$, and there it is [17, Thm. 1.4] weakened by one substitution. Zhang computes $\ell(n,\{b\}) = \frac{s}{bn}\max_u\big(\frac{1}{F(u)}+\frac{1}{F(b-u)}\big)$ with $F(x)$ the largest $i \le n-1$ congruent to x modulo b; for x coprime to b that representative is the unique one in $[s, n-1]$, so $\min_u F(u)$ is exactly our j_0, and his $I(n,r)$ is our j_0. Replacing $F(b-u)$ by the cruder $n-1$ gives Proposition 4.2, which is why ours is an inequality where his is an identity. A reader wanting the sharp statement in this regime should take it from [17]; we keep the weaker form only because it is what the Lean development proves.

Remark 4.4. Proposition 4.1 is a lower bound only, which is all that is needed below. Zhang [17, Thm. 1.4] computes $\ell(n,\{b\})$ exactly, and in the regime $2b \le n-1$ his value agrees with ours, so the bound is sharp there.

Proof of Theorem 1.6. By Lemma 1.5 take $M = \{b\}$. If $b = 1$ then $\gcd(b,n) = 1$ and $j_+ = 1 = b \in M$ for $a = 1$, so Theorem 1.7 gives $\ell \ge 1/(n(n-1))$, which is stronger than every claim below. Assume $b \ge 2$ and apply Proposition 4.1. The two regimes are separated by $2b$ against $n-1$, not by b against $n/2$.

(i). If $2b > n-1$ then $\ell \ge s/(nbj_0) \ge 1/(n\cdot n\cdot n) = 1/n^3$, using $s \ge 1$, $b < n$ and $j_0 \le n-1$. If $2b \le n-1$ then $s = n-b \ge b+1$, so $j_0 \le s+b-1 \le 2s-2$, giving $2s-j_0 \ge 2$ and

$$\ell \;\ge\; \frac{2s-j_0}{2nbj_0} \;\ge\; \frac{1}{nbj_0} \;\ge\; \frac{2}{n(n-1)^2} \;\ge\; \frac{1}{2n^3},$$

using $b \le (n-1)/2$ and $j_0 \le n-1$, and $4n^2 \ge (n-1)^2$ for the last step.

(ii). Suppose $2b > n-1$, so $\ell \ge s/(nbj_0)$. Let $P = P(b)$ and let P^m be the least power of P, with $m \ge 0$ allowed, satisfying $P^m \ge s$. Then $P^m < Ps$ (for $s = 1$ take $m = 0$) and $\gcd(P^m, b) = 1$, so $j_0 \le P^m < Ps$ and $\ell > s/(nb\cdot Ps) = 1/(nbP) > 1/(n^2P)$. It remains to check $P < 2\log b + 2$. If $P \ge 13$ then every prime at most 11 divides b, so $b \ge e^{\theta(P-1)} > e^{(P-1)/2}$,

where $\theta(x) = \sum_{p \le x} \log p$ and $\theta(x) > x/2$ for $x \ge 11$ [13]; hence $P < 2\log b + 1$. For $P \le 11$ the least admissible b is $3, 2, 6, 30, 210$ for $P = 2, 3, 5, 7, 11$, and since $2\log b + 2$ increases with b it suffices to check those: the values are $4.20, 3.39, 5.58, 8.80, 12.69$, each larger than P. If instead M contains an element coprime to n, Theorem 1.7 gives $\ell \ge 1/(n(n-1)) \ge 1/(n^2(2\log n + 2))$.

(iii). By (ii) the cases $2b > n-1$ and $\gcd(b,n) = 1$ are done, so there remains $2b \le n-1$ with $\gcd(b,n) > 1$, where $s \ge b+1$. Let $g(b)$ be Jacobsthal's function, the largest gap between consecutive integers coprime to b; then $j_0 \le s + g(b) - 1$. By Iwaniec [7], $g(b) \ll \log^2 b$, so there is N_0 with $g(b) \le s/2$ for all $n \ge N_0$ in this range. For such n, $2s - j_0 \ge s/2$ and $j_0 \le 3s/2$, so $\ell \ge \frac{s/2}{2nb \cdot 3s/2} = \frac{1}{6nb} \ge \frac{1}{3n^2}$. For $3 \le n < N_0$, part (i) gives $\ell \ge 1/(2n^3)$, hence $\ell\, n^2 \log n \ge \log 3/(2N_0)$. Taking c to be the least of $1/3$, $\log 3/(2\log 3 + 2)$ and $\log 3/(2N_0)$ proves (iii); the second of these is the constant obtained from (ii), since converting $1/(n^2(2\log n + 2))$ to the shape $c/(n^2 \log n)$ requires $c \le \log n/(2\log n + 2)$, whose infimum over $n \ge 3$ is $\log 3/(2\log 3 + 2) = 0.2617\ldots$. The dependence on N_0, hence on Iwaniec's implied constant, is why c is not explicit. □

5. Localization of the uncovered region

This section is expository. For $M = \{r\}$ the localization below is in [17, §2: Lemmas 2.1 and 2.2 and the first paragraph of the proof of Lemma 2.4; the case $2r > n-1$ is Lemma 2.5], version 2: Dirichlet's theorem with $N = n-1$ forces $\|rt\| \le 1/n$ on $U(n,r)$, whence $U(n,r) \subseteq \bigcup_p [p/r - 1/(rn),\, p/r + 1/(rn)]$, coprimality reduction included. Replacing $\{r\}$ by M changes nothing in that argument. Bohman and Peng [2, §3] use the same idea constructively, taking a witness time whose denominator is built from deleted speeds. We note also that Dirichlet is not needed for the containment: since $\mathrm{LR}([n-1]) = 1/n$, the set on which every $j \in [n-1]$ has $\|jt\| \ge 1/n$ is the finite set $\{a/n : \gcd(a,n) = 1\}$ (see the opening line of the proof of [8, Prop. 10.4]), so off a finite set $U(n,M) \subseteq \bigcup_{b \in M} B_b$ in one line, which also gives Corollary 1.11 directly. The measure of the uncovered region as an object of study is [4], and measure together with component count already appears in [5, Lemma 1]. We record all of it here because the formalization and Section 7 consume it.

Sections 3 and 4 bound $\ell(n,M)$ from below by exhibiting an interval inside $U(n,M)$. This section determines $\ell(n,M)$ exactly. The mechanism is Dirichlet's approximation theorem: it forces every uncovered time to sit close to a fraction whose denominator was deleted, and there is nowhere else for the uncovered region to be.

Proof of Theorem 1.10. Dirichlet's approximation theorem with $N = n-1$ gives an integer q with $1 \le q \le n-1$ and $\|qt\| \le 1/(N+1) = 1/n$. If q were present the hypothesis would force $\|qt\| > 1/n$, so $q \in M$. Taking a nearest to qt turns $\|qt\| \le 1/n$ into $|t - a/q| \le 1/(qn)$.

For coprimality, write $d = \gcd(a,q)$, $a = da'$, $q = dq'$. Then $|t - a'/q'| = |t - a/q| \le 1/(qn) \le 1/(q'n)$, so $\|q't\| \le 1/n$ with $1 \le q' \le n-1$, and the same argument puts q' in M. □

The first consequence bounds the size of the uncovered region, uniformly in which speeds were deleted and how large they are.

Proof of Corollary 1.11. The complement of $U(n,M)$ in $[0,1)$ is a finite union of open intervals, so $U(n,M)$ differs from its interior by a finite set. Every interior point satisfies $\|jt\| > 1/n$ for all present j: if some present j had $\|jt\| = 1/n$ then t would lie on the boundary of B_j and hence not be interior to the complement of B_j. Theorem 1.10 therefore covers the interior by the intervals $[a/q - 1/(qn),\, a/q + 1/(qn)]$ with $q \in M$ and $0 \le a < q$. There are at most $\sum_{q \in M} q$ of them, of length $2/(qn)$ each, for a total of $\sum_{q \in M} q \cdot 2/(qn) = 2|M|/n$. □

The bound is of the right order: over all M with $|M| = k$ for $k = 1, 2, 3$ and $n \in \{16, 22, 28, 34, 40\}$ the largest value of $\lambda(U(n,M))/(2k/n)$ lies between 0.5139 and 0.6492. The maximiser is unique

in each of the fifteen cells and is usually a set of primes just above $n/2$, though not always: at $n = 16$, $k = 2$ it is $\{9, 11\}$, at $n = 16$, $k = 3$ it is $\{9, 11, 13\}$, at $n = 28$, $k = 3$ it is $\{1, 17, 19\}$, and at $n = 40$, $k = 3$ it is $\{23, 25, 29\}$.

The second consequence is an exact formula. The point of Theorem 1.10 is not only that the uncovered region sits near the deleted denominators, but that each of its components runs all the way back to one of them, so that the component is determined by an explicit minimum.

Definition 5.1. Let $b \in M$ and let a satisfy $0 \le a < b$ and $\gcd(a, b) = 1$. For present j write $r_j = ja \bmod b$, and set

$$\delta^+ = \min_{j \text{ present}} \begin{cases} \dfrac{(b - r_j)/b - 1/n}{j}, & b \nmid j, \\ \dfrac{1 - 1/n}{j}, & b \mid j, \end{cases} \qquad \delta^- = \min_{j \text{ present}} \begin{cases} \dfrac{r_j/b - 1/n}{j}, & b \nmid j, \\ \dfrac{1 - 1/n}{j}, & b \mid j, \end{cases}$$

$$\delta^0 = \max\Big\{ \tfrac{1}{jn} \ : \ j \text{ present}, \ b \mid j \Big\},$$

with $\delta^0 = 0$ when b has no present multiple. The *window* at a/b is

$$w(n, b, a, M) = \begin{cases} \delta^+ + \delta^-, & \delta^0 = 0, \\ \max\{\delta^+, \delta^-\} - \delta^0, & \delta^0 > 0, \end{cases}$$

read as 0 if the right-hand side is negative.

Theorem 5.2 (Proved; Exact formula)**.** *For every $n \ge 3$ and every M with $\emptyset \ne M \subsetneq [n-1]$,*

$$\ell(n, M) \ = \ \max_{b \in M} \ \max_{\substack{0 \le a < b \\ \gcd(a,b) = 1}} \ w(n, b, a, M).$$

The hypothesis $M \ne [n-1]$ is needed: at $M = [n-1]$ there is no present speed, $U(n, M) = [0, 1)$ and $\ell = 1$, while δ^+ and δ^- are minima over an empty family. Every M arising from a k-speed modification satisfies $2k < n$ and so is covered.

Proof. The maximum is attained inside $U(n, M)$. Fix b and a and let $t = a/b + \delta$ with $\delta^0 \le \delta \le \delta^+$. If $b \mid j$ then $j\,a/b \in \mathbb{Z}$, so $\|jt\| = \|j\delta\|$; here $j\delta \ge j\delta^0 \ge 1/n$ by the definition of δ^0, and $j\delta \le j\delta^+ \le 1 - 1/n$ by the second branch of δ^+. If $b \nmid j$ then $jt \equiv r_j/b + j\delta$, which is at least $r_j/b \ge 1/b > 1/n$ above the integer below it, and at most $1 - 1/n$ by the first branch. So $[a/b + \delta^0,\ a/b + \delta^+] \subseteq U(n, M)$, and symmetrically on the left; when $\delta^0 = 0$ the two sides join at a/b.

Every component of positive length is such a window. Let C be a component of $U(n, M)$ with $|C| > 0$, which exists by Theorem 1.6(i), and let t be an interior point of C, so $\|jt\| > 1/n$ for every present j. Theorem 1.10 gives $b \in M$ and a coprime to b with $|t - a/b| \le 1/(bn)$; assume $t \ge a/b$, the other case being the mirror image, and put $D = t - a/b$. Since $j \le n - 1$ we have $jD \le (n-1)/(bn) < 1/b$ for every present j.

First, $\delta^0 \le D$. Indeed if $b \mid j$ for a present j then $\|jt\| = \|jD\|$, and $jD < 1/b \le 1$, so $\|jt\| > 1/n$ forces $jD > 1/n$, that is $D > 1/(jn)$; taking the maximum over the present multiples gives $D \ge \delta^0$.

Now let $\delta^0 \le d \le D$ and $s = a/b + d$. If $b \nmid j$, write $ja = Qb + r_j$ with $0 < r_j < b$; then $js = Q + r_j/b + jd$. Its distance down to Q is $r_j/b + jd \ge r_j/b \ge 1/b > 1/n$, and its distance up to $Q + 1$ is $1 - r_j/b - jd$, which decreases in d and so is at least $1 - r_j/b - jD$. Now $r_j/b + jD < (b-1)/b + 1/b = 1$ rules out a wrap, so $1 - r_j/b - jD$ is one of the two distances from jt to $\mathbb{Z}$ and hence at least $\|jt\|$, which is at least $1/n$ because $t \in U(n, M)$. If instead $b \mid j$ then $\|js\| = \|jd\|$, with $jd \ge j\delta^0 \ge 1/n$ and $jd \le jD \le (n-1)/(bn) \le 1 - 1/n$.

So no present speed blocks between $a/b+\delta^0$ and t, whence $[a/b+\delta^0, t] \subseteq C$. It remains to identify both endpoints of C. On the right, the containment step already gives $[a/b+\delta^0, a/b+\delta^+] \subseteq U(n,M)$, and at δ^+ the minimum defining it is attained by some present j, for which $\|j(a/b+\delta)\| < 1/n$ for every δ slightly larger; so C ends at $a/b+\delta^+$. On the left, if b has a present multiple j then $\|j(a/b+\delta)\| = \|j\delta\| < 1/n$ for $0 \le \delta < 1/(jn)$, so C starts at $a/b+\delta^0$ and $|C| = \delta^+ - \delta^0$; the mirror argument bounds the component on the left of a/b by $\delta^- - \delta^0$, and w takes the larger of the two. If b has no present multiple then $\delta^0 = 0$, the two sides join at a/b, and the same argument at $-\delta^-$ gives $C = [a/b-\delta^-, a/b+\delta^+]$, of length $\delta^+ + \delta^-$. Either way $|C| = w(n,b,a,M)$, and taking C of maximal length gives the reverse inequality. □

Remark 5.3. Since the components are disjoint, $\lambda(U(n,M))$ is the sum of their lengths. It is *not* the case that every component is a window: $U(n,M)$ also has degenerate components, isolated points of the equality locus where some present j attains $\|jt\| = 1/n$ exactly. At $n=4$, $M=\{2\}$ there are three components and one window. Over all 65 504 nonempty $M \subsetneq [n-1]$ with $3 \le n \le 16$, degenerate non-window components occur in 43 849 cases. They carry no measure, so what is true, and what Theorem 5.2 gives, is that every *positive-length* component is a window; that holds with no exception over the same 65 504 cases. Hence $\lambda(U(n,M))$ is the sum of the windows *distinct as intervals*: the deduplication is load-bearing, and summing over base points without it is wrong in 62 816 of those 65 504 cases, because a base point whose b has a present multiple contributes the two windows $[a/b+\delta^0, a/b+\delta^+]$ and $[a/b-\delta^-, a/b-\delta^0]$ while a component may abut two different base points. This is bookkeeping, not a theorem, and the pairing of measure with component count is already in [5, Lemma 1]. Verified on all 7 315 pairs with $|M| \le 3$, $n \in \{12, 16, 20, 24, 28\}$.

Three points about the formula. It costs $O\big(n \sum_{b\in M} \varphi(b)\big)$ arithmetic operations, against $O(n^2 \log n)$ for sweeping the endpoints $(in \pm 1)/(jn)$ directly; for a single deleted speed b of order n that is a saving of the logarithmic factor only, and the saving is real only when the deleted speeds are small. It excludes *all* of M from the minima, which is what Lemma 1.5 cannot do: monotonicity replaces M by a single $\{b\}$ and so still sees the other deleted speeds as present. And it turns Theorem 3.1 into a finite, exactly checkable condition.

Corollary 5.4 (Proved; Unconditional criterion, explicit form)**.** *Let $n \ge 3$ and $k \ge 1$ with $2k < n$, and let $M \subseteq [n-1]$ with $|M| = k$. If some $b \in M$ and some a with $0 \le a < b$ and $\gcd(a,b) = 1$ satisfy*

$$w(n,b,a,M) \;>\; \frac{2k}{n(n-2k)},$$

then $\mathrm{LR}(([n-1] \setminus M) \cup R) \ge 1/n$ for every admissible R.

Proof. Immediate from Theorem 5.2 and Theorem 3.1. □

Corollary 5.4 is stronger than Corollary 3.2, and by Theorem 5.2 it is equivalent to the criterion of Theorem 3.1 itself, so it exhausts what the covering bound can reach. The table below is therefore a consistency check on the implementation rather than independent evidence; we record it because it is what the code computes. Counted over *all* k-subsets M, at $n = 22, 26, 30, 34, 38, 42$:

n	22	26	30	34	38	42
$k=1$	38.10	36.00	41.38	42.42	37.84	41.46
$k=2$	21.43	19.33	23.89	21.78	24.77	22.44
$k=3$	14.44	13.13	11.88	18.37	16.82	15.55

The same agreement holds at $n = 20, 24, 28, 32, 36, 40$ for $k = 1, 2, 3$, and the identity $\ell(n, M) = \max_{b,a} w(n, b, a, M)$ of Theorem 5.2 was checked directly, in exact rational arithmetic, on all 134 721 pairs (n, M) with $|M| \le 4$ and $n \in \{16, 20, 24, 28, 32, 36\}$.

Remark 5.5. Condition (c) of Corollary 3.2 is empty for every $k \ge 2$. In that regime $2b \ge n$, and $s \le j_0 \le n - 1$, so

$$s(n - 2k)(j_0 + n - 1) \;\le\; j_0(n - 4) \cdot 2(n - 1) \;<\; 2n\, j_0(n - 1) \;\le\; 2kb\, j_0(n - 1),$$

the last step using $k \ge 2$ and $2b \ge n$. Condition (c) therefore contributes only at $k = 1$, where it fires 331 291 times for $n \le 2000$; for $2 \le k \le 10$ and $n \le 2000$ it fires not at all. This is why the $k = 2$ and $k = 3$ figures for conditions (a)–(c) coincide with those for (a)–(b) alone. Corollary 5.4 is not subject to this: it fires at every k for which the criterion does.

6. The coprime case

Proof of Theorem 1.7. Since $1 \le a < n$ we have $a/n - \frac{1}{n(n-1)} \ge 0$ and $a/n \le \frac{n-1}{n} < 1$, so the interval lies in $[0, 1)$. Let $j \in [1, n)$ with $j \ne j_+$, let $\varepsilon \in [0, \frac{1}{n(n-1)}]$ and put $t = a/n - \varepsilon$. Write $r = ja \bmod n$. As $\gcd(a, n) = 1$ and $1 \le j < n$ we have $n \nmid ja$, so $r \ne 0$; and $r = 1$ would give $j \equiv a^{-1}$, that is $j = j_+$, so $r \ne 1$. Hence $2 \le r \le n - 1$. Also $j\varepsilon \le (n - 1) \cdot \frac{1}{n(n-1)} = \frac{1}{n}$. Writing $ja = nq + r$,

$$jt = q + \Big(\frac{r}{n} - j\varepsilon\Big), \qquad \frac{r}{n} - j\varepsilon \in \Big[\frac{r-1}{n}, \frac{r}{n}\Big] \subseteq \Big[\frac{1}{n}, 1 - \frac{1}{n}\Big],$$

and a real number in $[1/n, 1 - 1/n]$ is at distance at least $1/n$ from every integer. □

The constant is attained, and for a reason rather than by coincidence.

Proposition 6.1 (Computed; lower bound proved for all odd $n \ge 5$; equality computed for $n \le 41$)**.** *For every odd $n \ge 5$, $\ell\big(n, \{(n-1)/2\}\big) \;\ge\; \dfrac{1}{n(n-1)}$, with equality for every odd n with $5 \le n \le 41$.*

Proof. Put $b = (n - 1)/2$, so $s = n - b = (n + 1)/2$. Since s and b differ by 1 they are coprime, so the least $j \ge s$ with $\gcd(j, b) = 1$ is $j_0 = s$ itself. Also $2b = n - 1 < n$, so the speed $2b$ is present and Proposition 4.1 applies, giving

$$\ell\big(n, \{b\}\big) \;\ge\; \frac{2s - j_0}{2nb\, j_0} \;=\; \frac{s}{2nb\, s} \;=\; \frac{1}{2nb} \;=\; \frac{1}{n(n-1)}.$$

Note also that b is coprime to n, since any common divisor of $(n - 1)/2$ and n divides $\gcd(n - 1, n) = 1$, so Theorem 1.7 gives the same lower bound independently.

The matching upper bound is not proved here. It is verified by exact rational computation of $\ell(n, \{b\})$ for every odd n with $5 \le n \le 41$, where equality holds in every case; the checker is committed. Proposition 12.1 gives only $\ell \le (n-2)/(n(n-1))$, which is larger by a factor $n - 2$, so the upper bound needs an argument the present method does not supply. □

Neither primality of n nor n being large plays any role. When n *is* prime, more is true: $b = (n - 1)/2$ is then the worst single deletion there is. Over all 76 primes with $5 \le n \le 400$ the minimum of $\ell(n, \{b\})$ over b equals $1/(n(n - 1))$ and is attained at $b = (n - 1)/2$, in every case; and $1/(n(n-1))$ bounds $\ell(n, \{b\})$ from below for every b, which is Theorem 1.7 again since every $b < n$ is coprime to a prime n. So at prime n the covering criterion of Theorem 3.1 can never fire at the extremal deletion: it would need $1/(n(n - 1)) > 2/(n(n - 2))$, that is $0 > n$.

For even n the extremal deletion is not so simply described. The value at $b = n - 2$ has the closed form $\ell(n, \{n-2\}) = \frac{2}{n(n-2)}\big(\frac{1}{P} + \frac{1}{n-2-P}\big)$, with P the least prime not dividing $n - 2$, exact

for all even $6 \le n \le 400$ except $n = 6, 8$; at $n = 32$ this reads $\frac{2}{960}(\frac{1}{7} + \frac{1}{23}) = \frac{1}{2576}$. But $b = n - 2$ is not always the minimiser: over the 148 even n with $6 \le n \le 300$ it is so in 121 cases and not in 27, the first exceptions being $n = 10, 14, 18, 24$.

7. Computations

All arithmetic is exact over $\mathbb{Z}$ and $\mathbb{Q}$ in `i128`; no floating point is load-bearing. The programs are `code/perturbation/perturb_*.rs` in the accompanying archive, the subcommands and the values each regenerates are documented in `code/perturbation/README.md`. Components of $U(n, M)$ are obtained by sorting the endpoints $(in \pm 1)/(jn)$ of the excluded intervals and sweeping. Each range is stated where it is used; they are not uniform.

A complete decision at $k = 2$. For $5 \le n \le 100$ and every M with $|M| = 2$ there is no admissible R at all: the conjecture holds for every 2-speed modification in that range. By Remark 1.2 that is exactly the statement that every set of $n - 1$ distinct positive integers with precisely two members at least n is lonely, for those n.

This is a decision rather than a sample, because two rigorous bounds make the enumeration finite. Theorem 1.4 gives $\min R \le 4/(\ell(n, M)(n-4))$, which bounds the smaller inserted speed; over $n \le 30$ the largest value taken is 136. Given v_1, the residual $U(n, M) \setminus B_{v_1}$ must lie inside B_{v_2}, and B_{v_2} is a *disjoint* union of open intervals, so each connected piece of the residual lies in a single one of them: a residual component of length L forces $v_2 < 2/(Ln)$. If the residual has no component of positive length it is a finite set of rationals p/q, and $\|v_2 p/q\| < 1/n$ depends only on v_2 modulo q, so one period of $\mathrm{lcm}(q)$ decides that branch; it did not arise.

Totals over $5 \le n \le 100$: 161 696 deletion sets, 7 520 944 pairs (M, v_1) examined and 13 989 171 enumeration steps, no covering R, and no branch truncated by a cap. The conjecture is known for at most 13 runners: seven in [3], eight in [12], nine and ten in [16], and eleven to thirteen in [14]. So 161 414 of those 161 696 sets lie beyond the range proved outright.

The enumeration does not carry over to $k = 3$ unchanged, and we record why. The second of the two bounds above is a connectivity argument: it reads off $v_2 < 2/(Ln)$ from the fact that a residual component must lie inside a single interval of B_{v_2}. That inference needs exactly one speed to remain. With two still to place, a residual component may be covered jointly by $B_{v_2} \cup B_{v_3}$ and neither speed need satisfy the bound. At $k = 3$ the recursion reaches a level with two speeds remaining, so the connectivity bound is outside its hypothesis there. The repair is to use Theorem 1.4 on the residual at every level where $r \ge 2$ speeds remain, giving $\min \le 2r/(\ell(n - 2r))$, which is uniform in r and is the same measure argument that removes the cap at level zero.

With that bound in place the enumeration at $k = 3$ decides $7 \le n \le 40$: 3 021 814 pairs (M, v_1) and 153 363 483 enumeration steps, no covering R, and no branch truncated by a cap. The corrected bound multiplies the number of steps by about seven relative to the connectivity version, which is the measure of how much of the space the latter was missing. The $k = 2$ decision is unaffected either way, since exactly one speed remains at every recursive level there.
Reproduction status. The $k = 2$ totals above are regenerated by `code/perturbation/perturb_coverk.rs 2 5 100`: the rows $5 \le n \le 100$ sum to 161 696 sets, 7 520 944 pairs and 13 989 171 steps with no cover and no capped branch (rows 31 to 66 come from a run of 17 October 2026, the others from 18 October 2026, same program). The $k = 3$ totals are the column sums of the log of the original run (`data/repro_logs/perturb_coverk_k3_7_40_original.log`); the archived program reproduced its rows for $7 \le n \le 22$ exactly, and the rows $23 \le n \le 40$ were not re-run. The $k = 3$ decision for $23 \le n \le 40$ is therefore a COMPUTED result of a development run that is not regenerated here.

Zhang [17] reports exact $k = 3$ searches inside windows (removals in the mid-range, $n \le 140$; see his Remark 6.3 and Section 8); we have not found a height-free decision for $k = 3$ in the literature read, and have not searched exhaustively. The case $k = 1$ is settled by [6, Thm. 2.3] together with [17, Thm. 1.2, Lemma 2.3] and a computation; [6, Thm. 2.4] gives finiteness for fixed r but is not effective, and [17, Thm. 1.4] computes the components in a sharper form than anything here. For $k = 2$, [17, Thms. 7.5, 7.10, 7.13] prove non-existence of tight two-speed modifications for the removed sets $\{2, r'\}$ $(n \ge 19)$ and $\{q, r'\}$ $(n \ge \max(40, 10q))$, and give a census for $8 \le n \le 140$ with inserted speeds at most $8n$; version 1 had verified $6 \le n \le 34$ under $\max R \le 4n$. The computation here is independent of those arguments: Theorem 1.4 bounds $\min R$ with no hypothesis on the inserted speeds, so the enumeration is finite without a cap, and it covers $5 \le n \le 100$ for $k = 2$ and $7 \le n \le 40$ for $k = 3$. For $k = 2$ it is not a stronger result than Zhang's theorems on the removed sets they cover.

As a control, the tight set $\{1, 4, 5, 6, 7, 11, 13\}$ at $n = 8$, due to Goddyn and Wong [6] and also recorded by Zhang, which is the 2-speed modification with $M = \{2, 3\}$ and $R = \{11, 13\}$, lies in the range swept here. It is tight, not a counterexample, and the search correctly reports no covering: the times $1/8$ and $7/8$ survive, since neither 11 nor 13 is divisible by 8.

The search is instrumented against the failure mode of a test that cannot succeed. Six positive controls check that the interval subtraction removes a contained interval, trims a partial overlap to the exact expected endpoint, leaves a disjoint interval untouched, splits a straddling interval in two, and that the congruence branch fires at $t = 1/4$ and $t = 1/2$, where a covering residue class exists. The negative control is that the search reports nothing for $n \le 13$, where the conjecture is a theorem.

7.1. **The tight 2-speed modifications, exhaustively.** The same enumeration, run to decide tightness rather than to look for counterexamples, classifies the tight 2-speed modifications outright. With no cap on the inserted speeds, over $1.41 \cdot 10^8$ enumeration steps and with no branch left undecided:

Proposition 7.1 (Computed)**.** *For $5 \le n \le 100$, the following three 2-speed modifications of $[n-1]$ are tight, and the enumeration finds no others:*

$$\{2,4,6,8\}\ (n=5), \qquad \{1,4,5,6,7,11,13\}\ (n=8), \qquad \big([73] \setminus \{70,72\}\big) \cup \{140,144\}\ (n=74).$$

Two caveats on completeness, both in the same direction. The search records, for each deleted pair and each smallest inserted speed, the first covering completion it finds rather than all of them, so a second tight set sharing both could in principle go unlisted; enumerating all completions instead changes nothing over $5 \le n \le 34$, where the raw hits collapse to the same two sets. And the interval arithmetic represents the complement of a closed bad set by closed intervals, so it retains endpoints touched exactly and thereby over-estimates the surviving region. Both errors can only cause a covering to be missed, never invented, so Proposition 7.1 should be read as listing the tight modifications the enumeration finds, with no counterexample to the conjecture anywhere in the range, rather than as a proof that no fourth tight set exists.

None of the three is new, and it is worth being precise about why. The first is the dilation $2 \cdot [4]$. The second is the Goddyn–Wong set above. The third is an instance of the simultaneous-acceleration theorem of Goddyn and Wong [6], stated as [10, Thm. 12]: accelerating r by a factor m within $[n-1]$ preserves tightness when r shares a factor with every element of $[n-r,\, m(n-r)-1]$, and accelerations at several r compose. Here $r = 70$ and $r = 72$ are both accelerated by $m = 2$, with intervals $[4, 7]$ and $[2, 3]$; since $70 = 2 \cdot 5 \cdot 7$ meets each of $4, 5, 6, 7$ and 72 meets each of $2, 3$, both qualify.

So the content of Proposition 7.1 is negative, and that is its interest. A complete characterisation of tight instances is open, and the converse of that theorem is known to fail in general

[10]. Within 2-speed modifications, up to $n = 100$, the converse holds: every tight instance is a dilation or a Goddyn–Wong acceleration, with no sporadic example.

Tightness of the three was re-checked independently of the search, by computing $\mathrm{LR}(V)$ in exact integer arithmetic over the candidate times a/q with q dividing some $|v_i \pm v_j|$ or $2v_i$. That routine returns $\mathrm{LR} = 1/4, 1/5, 1/6$ on $\{1,2,3\}$, $\{1,2,3,4\}$, $\{1,3,4,5,9\}$, and returns $2/13$ on a perturbation of the $n = 8$ set, so it distinguishes tight from non-tight rather than reporting tightness unconditionally.

Remark 7.2 (The closest unconditional result on this object)**.** For large n, Bohman and Peng [2, Thm. 1.3], with the range improved by Pomerance [11], show that a set with $n < \max R \le 2n - c(\log n)^2$ is not tight. In the notation here that settles every k-speed modification, for every k, whose inserted speeds all lie below $2n - c(\log n)^2$; their witness time is again a fraction whose denominator is assembled from deleted speeds. It is asymptotic and ineffective in n, so it does not decide any particular small n, which is what Section 7 does; but it is the strongest existing unconditional statement about the objects of this paper.

8. The divisibility obstruction

Covering a rational point of the uncovered region is not an analytic condition but a divisibility one, and the two families of rational points that always lie in $U(n, M)$ impose competing demands.

This section is expository: the content is known. Proposition 8.1 is the mechanism behind [17, Lemma 2.3], which is in turn immediate from [6, Lemma 2.2], and Proposition 8.2 is the same observation at the base point $1/n$. We include the statements because the Lean development uses them and because they are what the search of Section 7 prunes on, not as new results.

Proposition 8.1 (Proved; Divisibility at a base point)**.** *Let $b < n$, $\gcd(a, b) = 1$, and let v satisfy $\|v\, a/b\| < 1/n$. Then $b \mid v$.*

Proof. Write $r = va \bmod b$, so $\|v\, a/b\| = \min(r, b - r)/b$. If $r \neq 0$ this is at least $1/b > 1/n$. Hence $r = 0$, that is $b \mid va$, and $\gcd(a, b) = 1$ gives $b \mid v$. □

Proposition 8.2 (Proved; The times j/n)**.** *Let $\gcd(j, n) = 1$. Then $j/n \in U(n, M)$ for* every *$M \subseteq [n-1]$, and any v with $\|v\, j/n\| < 1/n$ satisfies $n \mid v$.*

Proof. A present speed q has $1 \le q < n$, so $n \nmid qj$ and $\|q\, j/n\| \ge 1/n$; this uses nothing about M. For the second claim, $\|v\, j/n\| < 1/n$ forces $vj \equiv 0 \pmod n$, and $\gcd(j, n) = 1$ gives $n \mid v$. □

Both are equivalences: $b \mid v$ makes a/b a centre of a bad interval of v, and likewise for $n \mid v$ and j/n. We verified them as such in exact arithmetic, by `code/perturbation/perturb_divis.rs`: 593 442 instances of Proposition 8.1 over $3 \le n \le 40$, $1 \le b < n$, a a unit mod b and $1 \le v \le 3n$; and 132 684 instances of Proposition 8.2 over $3 \le n \le 60$ with $1 \le v \le 3n$. No exception. The membership $j/n \in U(n, M)$ was checked over *all* 2^{n-1} subsets M for $3 \le n \le 14$, including $M = [n-1]$: 121 368 instances, no exception.

Corollary 8.3 (Proved; Necessary conditions on R; cf. [17, Lemmas 2.3, 6.1])**.** *Suppose $\mathrm{LR}(([n-1] \setminus M) \cup R) < 1/n$. Then some $v \in R$ is divisible by n; and for every $b \in M$ having no present multiple, some $v \in R$ is divisible by b. Writing M_0 for the set of such b, if $|M_0| = k$ then the k speeds of R must realise the $k+1$ demands $M_0 \cup \{n\}$, so some $v \in R$ is divisible by $\mathrm{lcm}(d, d')$ for two distinct demands d, d'.*

The statement differs from [17, Lemma 2.3] in three ways: he assumes $2r > n - 1$ where the condition here is the weaker "no present multiple" (which $2b \in M$ also satisfies), he assumes tightness where we assume $\mathrm{LR} < 1/n$, and the clause about $n \mid v$ and the pigeonhole over

$M_0 \cup \{n\}$ are not in the statements we read (his Lemma 6.1 extends the divisibility to every removed $r < n$ under tightness). The mechanism is his.

Proof. If b has no present multiple then $\delta^0 = 0$, so $a/b \in U(n, M)$ by Theorem 5.2; $j/n \in U(n, M)$ always by Proposition 8.2. Each must be covered by some $v \in R$, and Propositions 8.1 and 8.2 identify the divisor. The last clause is pigeonhole on $k+1$ demands and k speeds. □

At $k = 1$ the two demands collide on a single speed, and that alone settles the regime.

Theorem 8.4 (Proved; Criterion D)**.** *Let $n \geq 3$, $k = 1$, and $M = \{b\}$ with $2b \geq n$, so that b has no present multiple. If*

$$\ell(n, \{b\}) \cdot n \cdot \operatorname{lcm}(b, n) \;\geq\; 2,$$

then $\mathrm{LR}(([n-1] \setminus \{b\}) \cup R) \geq 1/n$ for every admissible R.

Proof. Suppose not, and let $R = \{v\}$. By Corollary 8.3, $b \mid v$ and $n \mid v$, so $\operatorname{lcm}(b, n) \mid v$ and in particular $v \geq \operatorname{lcm}(b, n)$. The longest component W of $U(n, M)$ is connected and contained in B_v, which is a disjoint union of open intervals, so W lies in one of them and $\ell(n, \{b\}) = |W| < 2/(vn)$. Hence $v < 2/(\ell n)$, contradicting the displayed hypothesis. □

Theorem 8.4 is not new, and we claim no priority for it. The regime $2b \geq n$ at $k = 1$ is already closed in the literature by exactly this route: [17, Lemma 2.3] gives $b \mid v$, after which [17, Thms. 1.1, 1.2] classify the single-speed modifications. We record the criterion in this form only because it is what the formalization proves and because the numerical comparison below locates the measure criterion's loss. Exhaustively over all b with $2b \geq n$, Theorem 8.4 certifies 100% of them at $n = 20, 26, 32, 38, 44, 50$, against $30.0, 30.8, 37.5, 31.6, 40.9, 40.0$ per cent for the criterion of Theorem 3.1, and 100% is also the truth. So on this regime the measure criterion is losing roughly 60 per cent of what divisibility sees, which is the useful reading. The mechanism is visible at $n = 32$, $M = \{30\}$: the covering bound caps v at 171, while $\operatorname{lcm}(30, 32) = 480$.

For $k \geq 2$ the two demands need not fall on the same speed, and Corollary 8.3 becomes a constraint rather than a contradiction: the speed covering the base point of the longest window is divisible by b, hence at least $b\lceil n/b \rceil$, and so covers only the radius $1/(vn)$ about a/b, leaving two arms to the remaining $k-1$ speeds. Each arm is connected, so the connectivity bound applies to it, and the analysis is the finite recursion of Section 7 rather than a closed criterion. What the divisibility does supply there is a sound and substantial pruning: by Proposition 8.2, if any j/n survives the speeds chosen so far, the last remaining speed must be a multiple of n.

Propositions 8.1 and 8.2, the membership $j/n \in U(n, M)$, and the divisibility half of Theorem 8.4 are formalized; see Section 7 and the table in the Formalization section.

- Theorem 1.7 was tested on all M of size 1 with $n \leq 44$ (602 instances with M meeting the units mod n) and of size 2 with $n \leq 34$ (4857 instances): no violation, and the value $1/(n(n-1))$ attained.
- The lower bound of Proposition 4.1, together with Theorem 1.7, was compared against the exhaustively computed $\ell(n, \{b\})$ for all 4004 pairs $b < n \leq 90$: no violation. Over that range $\min \ell(n, \{b\})\, n(n-1)P(b) = 1.59563$, at $(n, b) = (74, 70)$, and the same minimum for the proved lower bound is 1.39048, at the same pair.
- Over all $b < n \leq 400$: the largest coprime gap $j_0 - s$ is 9, at $(n, b) = (212, 210)$; and in the regime where it is used, namely $2b \leq n - 1$, the quantity $(2s - j_0)/s$ is always at least 0.625, attained at $(n, b) = (14, 6)$, with no case where $2s - j_0 \leq 0$.
- $P(b) < 2 \log b + 2$ was verified for all $2 \leq b \leq 2 \cdot 10^5$, consistent with the case check in the proof of Theorem 1.6(ii).

The logarithm in Theorem 1.6(iii) is not an artifact of the proof, but establishing that needs an *upper* bound on ℓ, which Proposition 4.1 does not provide. Zhang [17, Thm. 1.4] computes

$\ell(n, \{b\})$ exactly; his formula at $M = \{n-2\}$ gives $\ell\, n(n-1) = 2/P(n-2) + o(1)$ as $n \to \infty$ along any sequence with $P(n-2)$ fixed, and $P(b) \to \infty$ along primorials, so no bound of the form c/n^2 holds for all M. We record the agreement of our computed values with that formula, without relying on it elsewhere: $P = 7$ for $b = 720$ gives limit $2/7 = 0.2857$ and measured 0.28892 at $n = 722$; $P = 11$ for $b = 420$ gives 0.18182 and measured 0.18715 at $n = 422$; $P = 13$ for $b = 2310$ gives 0.15385 and measured 0.15478 at $n = 2312$.

9. Formalization

The covering bound, the coprime component bound, the localization theorem and the unconditional criterion are formalized in Lean 4 with Mathlib. Each declaration builds with no `sorry` and depends only on `propext`, `Classical.choice` and `Quot.sound`. The development comprises 70 theorems and 5 definitions, across the fifteen modules `Perturb*.lean`, `CoveringBound.lean` and `PhaseBound.lean`; the count is reproduced by `grep -cE '^(theorem|def)'` on those files.

Statement	Lean declaration	constant
Cor. 1.8, coprime case	`perturbation_bound_of_coprime`	$(2+4/n)k$
Thm. 1.4, assembled	`covering_bound`	$(2+4/n)k$
Lemma 2.1	`volume_badSet_inter_Icc`	$(2+4/n)/(vn)$
B_v as a union of intervals	`badSet_eq`	n/a
Thm. 1.4, arithmetic core	`covering_forces_small_speed`	general
Thm. 1.7	`perturb_component_of_coprime`	sharp
Thm. 1.7, core	`perturb_component`	sharp
uniqueness of j_+	`residue_one_unique`	n/a
$r \neq 0$	`residue_ne_zero`	n/a
distance to $\mathbb{Z}$ on $[1/n, 1-1/n]$	`dist_int_ge_of_mem_Icc`	n/a
Lemma 1.5	`uncovered_mono`	n/a
Thm. 1.7 in region form	`Icc_subset_uncovered`	n/a
Prop. 4.1, containment	`Icc_subset_uncovered_missing`	sharp
Prop. 4.1, upper edge	`hnon_of_j0`	sharp
Cor. 1.8, missing-speed case	`perturbation_bound_missing`	$(4+8/n)k$
maximality of j_0	`le_of_congr_of_ge`	n/a
j_0 exists in the window	`exists_coprime_window`	n/a
existence of a	`exists_a_of_coprime`	n/a
maximality, packaged	`hj0max_of_spec`	n/a
Thm. 1.6(i), two regimes	`exists_interval_missing`	explicit
Thm. 1.6(i)	`exists_interval_uncovered`	explicit
Thm. 3.1, covering step	`not_covered_of_long_interval`	$(2+4/n)k$
Thm. 3.1	`exists_lonely_time`	$(2+4/n)k$
Cor. 3.2, window from j_0	`Icc_subset_uncovered_of_j0`	sharp
Cor. 3.2, criterion	`lonely_of_j0_criterion`	$(2+4/n)k$
Cor. 3.2(a)	`lonely_of_one_mem`	$n > (6+8/n)k$
Cor. 3.2(b)	`lonely_of_criterion_b`	$(4+8/n)kbj_0$
Cor. 3.2(c), one-sided	`lonely_of_criterion_c`	vacuous
Prop. 4.2, left edge	`missing_nonmultiple_protects_left`	sharp
Prop. 4.2	`Icc_subset_uncovered_twosided`	sharp
Cor. 3.2(c)	`lonely_of_criterion_c'`	$(2+4/n)kbj_0j_1$
Thm. 1.10	`exists_deleted_fraction`	sharp
Thm. 1.10, denominator form	`exists_deleted_denominator`	sharp
Thm. 1.10, lowest terms	`deleted_denominator_of_dvd`	sharp
Cor. 1.11, covering step	`uncoveredStrict_subset_union`	sharp
Thm. 5.2, abutment	`uncovered_between`	sharp
uncovered $\to$ strict on an interval	`uncoveredStrict_of_Icc_subset`	sharp
Thm. 5.2, abutment left	`uncovered_between_left`	sharp
two-sided window, any regime	`Icc_subset_uncovered_twosided_general`	sharp
two-sided containment, per speed	`Icc_subset_uncovered_twosided_direct`	sharp
Cor. 5.4, two-sided	`lonely_of_window_twosided`	$(2+4/n)k$
Cor. 1.11 on a component	`exists_base_point_of_Icc`	sharp
Cor. 5.4	`lonely_of_window`	$(2+4/n)k$
Prop. 8.1	`dvd_of_covers_base`	sharp
Prop. 8.2, membership	`mem_uncovered_of_coprime`	sharp
Prop. 8.2	`dvd_of_covers_trivial`	sharp
Thm. 8.4, divisibility half	`exists_survivor_of_not_lcm_dvd`	sharp

What is and is not machine-checked. The list below is the single authoritative inventory; the abstract and the table agree with it, and the table lists every declaration referred to here.

- **Formalized:** Lemma 2.1; Theorem 1.4; Lemma 1.5; Theorem 1.7; Proposition 4.1; Proposition 4.2; the selection of j_0 and of a; Theorem 1.6(i); Theorem 3.1; Theorem 1.10; Propositions 8.1 and 8.2 in full, and the divisibility half of Theorem 8.4; the abutment half of Theorem 5.2 and its containment half in the restricted form described below; Corollary 5.4; Corollary 3.2 in all three cases; and *both* displays of Corollary 1.8.
- **With a weaker constant.** The formalized strip lemma is proved by a union bound over the translates meeting the window rather than by periodicity. The translates that can meet $[x, x+\ell]$ are indexed by the integers in an open interval of length $v\ell + 2/n$, of which there are at most $v\ell + 1 + 2/n$, and each contributes at most $2/(vn)$; so Lean proves
$$\lambda\big(B_v \cap [x, x+\ell]\big) \;\le\; \frac{2\ell}{n} + \frac{2 + 4/n}{vn}$$
in place of the sharp $2\ell/n + 2/(vn)$. Everything downstream inherits the constant $2+4/n$ rather than 2. What Lean proves is the second display of Theorem 1.4, the bound on $\min R$; the first display, the inequality for $\sum_{v\in R} 1/v$, appears in the development only as a hypothesis of `covering_forces_small_speed` and is not itself formalized. The two displays of Corollary 1.8 are formalized in the forms $\min R \le (4+8/n)k\,n^3/(n-2k)$ and $\min R \le (2+4/n)k\,n(n-1)/(n-2k)$. Theorem 3.1 is formalized with $(2+4/n)k$ in place of $2k$; the loss is a factor $(n+2)/n$, and reaching 2 exactly would need the periodicity argument.
- **What that costs in reach.** The machine-checked form of Corollary 5.4 requires $w(n, b, a, M) > (2n+4)k/(n^2(n-2k))$ rather than $2k/(n(n-2k))$, and it is not the window w of Definition 5.1 that `lonely_of_window` consumes but a smaller one: its hypothesis $\delta_0 \ge 0$ makes the window one-sided, so it cannot straddle the base point. Its offset and far cap are now the sharp ones, $\delta_0 \ge 1/(jn)$ and $\delta_1 \le 1/n$; the crude $1/(2bn)$ and $1/(nb)$ survive only one level up, in `Icc_subset_uncovered_of_j0`, and it is there that the 4358 base points with $\delta^+ > 1/(nb)$, and the 488 at which b has a present multiple and $\delta^0 < 1/(2bn)$, apply. The figures below are for the union of `lonely_of_window`, `lonely_of_criterion_c'` and `lonely_of_one_mem`. Counted exhaustively over all k-subsets at $n = 22, 26, 30, 34, 38, 42$, what Lean certifies is $38.10, 32.00, 37.93, 42.42, 35.14, 39.02$ per cent at $k = 1$; $11.90, 18.33, 16.50, 20.83, 19.52, 21.83$ at $k = 2$; and $14.29, 12.17, 11.82, 17.80, 16.74, 15.43$ at $k = 3$, against the criterion's own $38.10, 36.00, 41.38, 42.42, 37.84, 41.46$; $21.43, 19.33, 23.89, 21.78, 24.77$, and $14.44, 13.13, 11.88, 18.37, 16.82, 15.55$.
- **Not assembled in Lean, and formalized only in part.** Three statements of this paper are not fully machine-checked, and we set out exactly what is missing. Theorem 5.2: both abutment halves are formalized, `uncovered_between` to the right and `uncovered_between_left` to the left, and `Icc_subset_uncovered_twosided_general` assembles them into a two-sided containment with no restriction on the regime, the hypothesis being that b has no present multiple rather than $n \le 2b$. The containment `Icc_subset_uncovered_missing` now carries the sharp per-speed offset $\delta_0 \ge 1/(jn)$ in place of $1/(2bn)$, and the far cap $\delta_1 \le 1/n$ in place of $1/(nb)$; `Icc_subset_uncovered_twosided_direct` gives the two-sided interval from per-speed conditions alone whenever b has no present multiple, and `lonely_of_window_twosided` is the resulting criterion. Together these lift the machine-checked reach at $k = 2$ from $10.00, 15.67, 14.04, 17.61, 16.07, 18.78$ to the figures above. What remains unformalized on Theorem 5.2 is the maximum over base points. Corollary 1.11: the covering statement is formalized, the summation $\sum_{q\in M} q \cdot 2/(qn) = 2|M|/n$ is not. Theorem 1.4: the first display is not formalized,

as noted above. Theorem 8.4: the divisibility half is `exists_survivor_of_not_lcm_dvd`, in full; the numerical assembly against the covering bound is not formed in Lean.

- **Not formalized:** parts (ii) and (iii) of Theorem 1.6, which rest on Chebyshev-type input and on Iwaniec's theorem respectively; Proposition 12.1; the elementary implication of Remark 3.3; the emptiness computation of Remark 5.5; and the recursion of Section 12. Remark 4.4 records a comparison and carries no formal content.

The row `lonely_of_criterion_c` records the superseded one-sided form of case (c) and is provably vacuous; `lonely_of_criterion_c'` replaces it and is not, though by Remark 5.5 it fires only at $k = 1$.

The Lean development depends on no external result beyond Mathlib. In particular Proposition 4.1 is proved there from scratch, Theorem 1.10 uses only Mathlib's `Real.exists_nat_abs_mul_sub_round_le`, and Remark 4.4 is used nowhere.

10. The covering bound away from modifications

The covering bound was set up for modifications of $[n-1]$, but nothing in it refers to that structure. Stated for an arbitrary family it reads as follows, and this form is machine-checked.

Proposition 10.1 (Verified; Lean: `covered_interval_short`)**.** *Let $v > 0$ and $0 < \delta \le \frac{1}{2}$. If $\|vt\| < \delta$ for every $t \in [x, y]$, then $(y - x)v < 2\delta$.*

The times at which a single speed fails to separate form intervals of length exactly $2\delta/v$, and an interval inside that union lies in one of them. The proof is the intermediate value theorem: otherwise vt takes the value $k + \delta$ somewhere in $[x, y]$, where the distance to the integers is exactly δ.

Corollary 10.2 (Verified; Lean: `le_ML_of_interval`)**.** *Let V be a finite family of nonzero integers, $0 < \delta \le \frac{1}{2}$, and j an index with $V_j > 0$. If some interval $[x, y]$ satisfies $\|V_i t\| \ge \delta$ for every $i \ne j$ and every $t \in [x, y]$, and $2\delta \le (y - x)V_j$, then $\mathrm{LR}(V) \ge \delta$.*

We record one application, to the four-speed loneliness spectrum. Kravitz [8] conjectured that LR of n speeds is either $s/(ns + 1)$ or at least $1/n$, proved it for $n \le 3$, and proved it for $n = 4$ and $n = 6$ under the lacunary hypothesis $v_n > 4v_{n-1}^4$ [8, Thm. 2.2]; the conjecture at $n = 4$ is false in general [9]. The first gap alone, that four speeds have $\mathrm{LR} = \frac{1}{5}$ or $\mathrm{LR} \ge \frac{2}{9}$, is open.

Write $\ell(v_1, v_2, v_3)$ for the longest interval on which all three of v_1, v_2, v_3 stay at distance $\ge \frac{2}{9}$. Corollary 10.2 applied with $\delta = \frac{2}{9}$ and $j = 4$ gives: if $\mathrm{LR}(v_1, \dots, v_4) < \frac{2}{9}$ then $v_4 < \frac{4}{9}/\ell$. Computation in exact rational arithmetic over all coprime triples with speeds at most 110 gives $\min \ell \cdot v_3 = \frac{1}{9}$, attained only at $(1, 2, 3)$, with second smallest value $\frac{4}{27}$, whence $v_4 < 4v_3$, so the first gap holds throughout $v_4 \ge 4v_3$. That range is strictly wider than the lacunary hypothesis $v_4 > 4v_3^4$ of [8, Thm. 2.2], since $4v_3^4$ exceeds $4v_3$ for $v_3 \ge 2$; the conclusion, however, is weaker, settling only the first gap and not the full spectrum statement. The minimum is a computation, not a theorem, so the constant 4 is HEURISTIC here.

What the bound does not do is close the remaining range, and the obstruction is sharp rather than technical. To force the contradiction $v_4 < v_3$ one needs $\ell v_3 > \frac{4}{9}$, whereas the median of ℓv_3 is $\frac{5}{9}$ and the small values are $\frac{1}{9}, \frac{4}{27}, \frac{2}{9}$. The triples realising them are, with few exceptions, exactly those with $v_3 = v_1 + v_2$. On such a triple $\|tv_3\| = \|tv_1 + tv_2\|$, so with $u = tv_1$ and $w = tv_2$ the three conditions collapse to a single condition on one point of $\mathbb{T}^2$, namely membership of $\{(u, w) : \|u\|, \|w\|, \|u + w\| \ge \delta\}$. At $\delta = \frac{2}{9}$ that set is two congruent right triangles of leg $1 - 3\delta = \frac{1}{3}$, of total area $(1 - 3\delta)^2 = \frac{1}{9}$, the same constant again. Deciding this case is a question about an orbit meeting a fixed planar region, which is the subject of the main lemma of [1]; we have not carried that out.

10.1. **A barrier for position-uniform covering lemmas.** The covering lemmas available for this kind of question all share a feature: their conclusion is that a group meets *every* rectangle of given dimensions, so the criterion they supply cannot depend on where the rectangle sits. That is what makes them convenient, and it is also what limits them here.

Fix $\delta = \frac{2}{9}$ and consider four speeds with $v_3 = v_1 + v_2$. Fixing $tv_2 \equiv \delta$ makes the second condition automatic and reduces the remaining three to asking whether a translated cyclic subgroup of $\mathbb{T}^2$ meets a rectangle. After normalising so that the subgroup is generated by $(1/n, c/n)$, the question becomes whether $\{kc \bmod n\}$, as k runs over an interval of L consecutive integers, meets an interval of length M; here n, c, L, M are exactly the data visible to a position-uniform criterion.

Proposition 10.3 (Computed)**.** *No criterion depending only on (n, c, L, M) decides this question. Explicitly, $(v_1, v_2, v_3, v_4) = (1, 6, 7, 11)$ and $(1, 6, 7, 17)$ both give $(n, c, L, M) = (6, 5, 2, 3)$, and the required t exists for the second and not for the first. The pairs $(1, 6, 7, 10)$ with $(1, 6, 7, 22)$, and $(1, 7, 8, 15)$ with $(1, 7, 8, 22)$, are further witnesses.*

Proof. Each assertion is a finite check on the four listed pairs: the two members of a pair agree on (n, c, L, M) by direct substitution, and the existence or non-existence of the required t is decided by scanning the finitely many candidate times, since an optimal t may be taken with denominator a sum or difference of two speeds. A criterion depending only on (n, c, L, M) assigns the same verdict to both members of a pair and is therefore wrong on one of them. □

The consequence is that a position-uniform covering lemma cannot decide every instance of this construction, whatever its constants. Measured on 147 099 instances, the best such criterion we found, namely that the largest gap of the point set is smaller than M, decides 94.17% of them and never errs; the residue is where instances agreeing on (n, c, L, M) disagree in outcome. This accounts for a ceiling that four separate arguments met at the same place, and it says what a proof must do instead: use the arithmetic of v_1, v_2, v_4, or a construction other than fixing $tv_2 \equiv \delta$.

10.2. **A Fourier criterion, and how it evades the barrier.** Proposition 10.3 bounds what a position-uniform covering lemma can do. The following criterion is not one, and is not subject to it.

Let $T \subseteq \mathbb{R}/\mathbb{Z}$ be a finite union of N intervals of total measure μ, let v be a positive integer, and write $F(k) = \int_T e^{-2\pi i k t}\, dt$. Say t is *good* if $\|vt\| \geq \delta$.

Proposition 10.4 (Proved; single mode)**.** *If no point of T is good, then $\operatorname{Re} F(v) \geq \cos(2\pi\delta)\, \mu$.*

Proof. If no point is good then $\|vt\| < \delta$ throughout T, so for each $t \in T$ the phase $2\pi vt$ lies within $2\pi\delta$ of a multiple of 2π, whence $\cos(2\pi vt) \geq \cos(2\pi\delta)$ by monotonicity of the cosine on $[0, \pi]$. Integrate over T. (Lean: `cos_lt_cos_of_nid_lt`, the pointwise step; the integration is not formalized.) □

Its contrapositive is a criterion: if $\operatorname{Re} F(v) < \cos(2\pi\delta)\mu$ then some point of T is good. Taking T to be the uncovered set of the three slower speeds and $v = v_4$, this decides 95.28% of the instances left open at four speeds, against 70.33% for the best criterion obtainable from component lengths alone.

Using all modes rather than one sharpens it. With g the indicator of the good phases,

$$\big|T \cap \{\|vt\| \geq \delta\}\big| \;=\; \hat{g}(0)\mu + 2\operatorname{Re} \sum_{m \geq 1} \hat{g}(m)\overline{F(mv)},$$

and truncating at M modes costs at most $2N/(\pi^2 vM)$, since $|F(k)| \leq N/(\pi|k|)$ for a union of N intervals and $|\hat{g}(m)| \leq 1/(\pi m)$. Positivity of the truncated sum less that tail forces a good

point. At $M = 40$ this decides every one of 49 121 instances except $(v_1, \dots, v_4) = (3, 4, 7, 11)$, where $\mathrm{LR} = \frac{2}{9}$ exactly, so the intersection has measure zero and no criterion demanding positive measure can decide it.

The reason this passes the barrier is that F depends on where T sits, not only on its translation-invariant data, so Proposition 10.3 does not apply. What remains, and is not proved here, is a lower bound for the truncated sum valid for every instance. Two structural facts bear on it. First, 1_T is a product of three indicators, $1_T(t) = h(v_1 t)h(v_2 t)h(v_3 t)$, so its Fourier coefficients are a convolution $F(k) = \sum_{av_1+bv_2+cv_3=k} \hat{h}(a)\hat{h}(b)\hat{h}(c)$, large only when k is close to a small integer combination of the slower speeds. Second, measurement of the margin across the same 49 121 instances shows it never falls below 0.372μ unless v_4 satisfies an integer relation with v_1, v_2 having coefficients at most 3; every instance with a small margin satisfies such a relation, and those instances are rank two, where a different and elementary argument applies. So the difficulty is confined to a case the criterion is not needed for. (The margin statement of this paragraph was measured in a session and its generating program is not archived; HEURISTIC. The counts 147 099, 94.17%, 49 121, 49 120 and the single undecided tuple $(3, 4, 7, 11)$ are regenerated by `code/perturbation/g1_gap_criterion.rs` and `r1_fourier_criterion.rs`, which use floating point; $\mathrm{LR}(3, 4, 7, 11) = 2/9$ is exact, `code/obstructions/mlcheck.rs`.)

11. Limits of the method

The covering bound is a measure estimate, and measure estimates on this problem are capped. We record where the cap sits, because it bounds what any sharpening of Section 3 can achieve.

First, at the baseline the bad sets already have full measure. By Dirichlet every t has some $q \le n-1$ with $\|qt\| \le 1/n$, so the set on which every $j \in [n-1]$ is protected is the finite set $\{a/n : \gcd(a, n) = 1\}$, of measure 0. Hence $\lambda\big(\bigcup_{j\in[n-1]} B_j\big) = 1$ exactly, which we verified in exact arithmetic for $n = 5, 7, 11, 13, 17, 23, 31, 41$. Any argument of the shape "the bad sets cannot cover because their measure is below 1" therefore fails at the extremal case. This is the classical factor-of-two obstruction, recorded in [15, Prop. 1.1] and sharpened there by the example of the primes in $(n/4, n/2)$, for which the union bound is tight to $1 + O(1/n)$.

Second, in the modification setting the inserted speeds have a measure surplus. The k speeds of R carry total bad-set measure exactly $2k/n$, while $\lambda(U(n, M)) \le 2k/n$ with the extremal ratio $\lambda(U)/(2k/n)$ lying between 0.5139 and 0.6492 over all M with $|M| \le 3$ and $n \in \{16, 22, 28, 34, 40\}$, and never reaching 1. So R always has between 35 and 49 per cent more measure than the region it must cover, and no refinement of the measure step can force non-covering.

Third, the criterion has an explicit ceiling. Its strength is governed by the smallest $\ell(n, \{b\})$, and that minimum behaves like $2/(n^2 P(n-2))$, with $P(m)$ the least prime not dividing m: at $b = n-2$ one has $s = 2$ and the binding unit is the least integer at least 2 coprime to $n-2$. Over $n = 50, 200, 800, 3200$ the ratio of $\ell(n, \{n-2\})\, n^2$ to $2/P(n-2)$ is 1.163, 1.036, 1.009, 1.002. The worst n are those with $n-2$ a primorial, where $\ell\, n^2 \log n$ stays near 1; this is why Theorem 1.6(ii) has the shape it does, and it is sharp.

The conclusion we draw is that the results of this paper are the ones the method reaches, and that progress on the conjecture itself will not come from sharpening the measure step. The modern advances for small numbers of runners [3, 12, 16, 14] are congruence sieves, a different mechanism.

12. Bounding every inserted speed

Theorem 1.4 bounds $\sum_{v\in R} 1/v$, hence $\min R$, but not every element of R. For $k = 1$ these coincide. For $k \ge 2$ one would like to recurse: fix the smallest inserted speed v_1, pass to

$U(n,M)\setminus B_{v_1}$, and apply Theorem 1.4 again with $k-1$ speeds. That needs a lower bound on the longest component of the residual region.

We tested the recursion with v_1 chosen adversarially to fragment the region, over all M with $|M|=2$, all $v_1\in[n,6n]$ and $8\le n\le 34$, in exact rational arithmetic. In every instance the residual region retained a component of positive length, and the minimum of $\big(\text{longest component of } U(n,M)\setminus B_{v_1}\big)\cdot n(n-1)$ was 0.375, attained at $n=8$, $M=\{2,6\}$, $v_1=24$, with no downward trend.

Proposition 12.1 (Proved; Ceiling)**.** *Let $n\ge 3$. For every $v\ge 1$, every connected subset of $[0,1)\setminus B_v$ has length at most $(n-2)/(nv)$.*

Proof. $[0,1)\setminus B_v$ is the disjoint union of the closed intervals $\big[\frac{i+1/n}{v},\ \frac{i+1-1/n}{v}\big]$, $0\le i<v$, each of length $(1-2/n)/v$. A connected set meeting two of them would meet the open gap between. □

There are thus two hard upper bounds on the longest component ℓ' of $U(n,M)\setminus B_{v_1}$: the ceiling $(n-2)/(nv_1)$ of Proposition 12.1, and $\ell(n,M)$ itself, since the residual is a subset of $U(n,M)$. Neither dominates: for v_1 large the ceiling is the smaller, while at $v_1\approx n$ the ceiling is about $1/n$ and $\ell(n,M)$ can be as small as order $1/n^2$, for instance at $M=\{n/2,n-1\}$ with n even. It is not always that small: $\ell(n,\{1,2\})=2/(3n)$ exactly, verified for $5\le n\le 40$.

Consequently a conjecture asserting a fixed fraction of $\ell(n,M)$ alone, with no dependence on v_1, is false: Proposition 12.1 forces $\ell'\to 0$ as $v_1\to\infty$ while $c\,\ell(n,M)$ stays fixed. A lower bound independent of v_1 fails because the ceiling tends to 0: at $n=8$, $M=\{2,6\}$ the longest component is $6.696\cdot 10^{-3}$ at $v_1=24$ and $1.144\cdot 10^{-5}$ at $v_1=65536$, the latter equal to the ceiling exactly, while the residual keeps positive measure: 0.031548 at $v_1=24$ and 0.054915 at $v_1=65536$, the latter approximately the limiting value $\frac{3}{4}\lambda(U(8,\{2,6\}))=123/2240=0.0549107$. Taking $v_1=n$ and minimising over M with $|M|=2$, the ratio $\ell'\big/\frac{n-2}{n^2}$ is $0.1000, 0.0900, 0.0653, 0.0455, 0.0383, 0.0333, 0.0294, 0.0263, 0.0238$ at $n=8,12,\dots,40$, so it decays like $1/n$. This is also what rules out a fixed fraction of the ceiling alone, and that we know only from data. The products $n\cdot$ ratio are $0.80, 1.08, 1.04, 0.91, 0.92, 0.93, 0.94, 0.95, 0.95$. The correct statement compares ℓ' with the smaller of the two bounds.

Conjecture 12.2 (Conjecture)**.** Let $k\ge 2$. There is $c=c(k)>0$, depending on k alone, such that for every n with $2k<n$, every M with $|M|=k$, and every nonempty set W of at most $k-1$ integers each at least n,

$$\ell_W\ \ge\ c\,\min\Big(\ell(n,M),\ \frac{n-2}{n\,\max W}\Big),$$

where ℓ_W is the length of the longest component of $U(n,M)\setminus\bigcup_{w\in W}B_w$.

The quantification over sets W, rather than over a single v_1, is what the recursion needs: at step j the region already has j punctures, and a statement about one puncture does not reach it. The ceiling is governed by $\max W$, since a connected subset of the complement of $\bigcup_{w\in W}B_w$ lies in the complement of each B_w and Proposition 12.1 applies to the largest.

Granting it, every inserted speed is bounded. Write ℓ_j for the longest component after j punctures. Theorem 1.4 applied to that region with the remaining $k-j$ speeds gives $v_{j+1}\le 2(k-j)/(\ell_j(n-2(k-j)))$, and each branch of the minimum is bounded:

$$v_{j+1}\ \le\ \frac{2(k-j)\,n\,\max_{i\le j}v_i}{c\,(n-2)\,(n-2k+2j)}\qquad\text{or}\qquad v_{j+1}\ \le\ \frac{1}{c}\cdot\frac{2(k-j)}{\ell(n,M)\,\big(n-2(k-j)\big)}.$$

The second is $1/c$ times the right-hand side of Theorem 1.4 with k replaced by $k-j$, so Theorem 1.6(i) bounds it by $4(k-j)n^3/(c(n-2k+2j))$. The first is bounded by induction, since $v_1\le 4kn^3/(n-2k)$ by Corollary 1.8. Iterating bounds every element of R and makes the

classification a finite computation for each n, answering for k-speed modifications a version of [17, Question 9.3]. Theorem 1.4 is stated for $U(n, M)$, but its proof uses only that the region is covered by the remaining B_v and has a component of length ℓ, so it applies verbatim.

The cost is not uniform in k. Step j multiplies the bound by $\frac{2(k-j)\,n}{c\,(n-2)(n-2k+2j)}$, which is about $2(k-j)/(cn)$ when $n \gg k$, and about $(k-j)/(cj)$ when $n-2k = O(1)$. The composed product $\prod_{j=1}^{k-1}$ of these is bounded by a constant times $c^{-(k-1)}$: numerically, at $n-2k=1$ and $c=0.2$, the product equals $0.907, 0.698, 0.516, 0.373$ times $c^{-(k-1)}$ at $k = 5, 10, 20, 40$. So the bound on $\max R$ is exponential in k through $c^{-(k-1)}$, with no additional factorial.

Evidence, in both regimes and at both puncture depths. With one puncture, over all M with $|M| = 2$ and all v_1 from n to $8n$, the ratio $\ell_W \big/ \min(\ell(n, M), (n-2)/(nv_1))$ has minimum 0.207341 over $21 \le n \le 44$, attained at $n = 32$, $M = \{18, 30\}$, $v_1 = 36$, and minimum 0.21875 over $8 \le n \le 20$, attained at $n = 8$, $M = \{3, 4\}$, $v_1 = 8$. With two punctures, exhaustively over all M with $|M| = 3$, all $9 \le n \le 22$ and all pairs $v_1 < v_2$ in $[n, 3n]$ (5 241 348 configurations), the minimum is $3/22 = 0.136364$, at $n = 13$, $M = \{5, 6, 7\}$, $W = \{13, 14\}$. In no instance did the region lose all positive-length components. The constant falls with puncture depth, consistent with the degradation just described.

Positivity alone is weaker than it looks: for $|M| = k \ge 2$ the set $([n-1] \setminus M) \cup W$ has at most $n-2$ speeds, so the Lonely Runner Conjecture predicts a floor strictly above $1/n$ for it, and strictness forces $\ell_W > 0$. So the positive-length half of Conjecture 12.2 is a consequence of the conjecture this paper is about; only the quantitative half is extra.

The restriction $|W| \le k-1$ is what does the work, not $k \ge 2$ as such. For $k = 1$ the residual can collapse to a finite set of isolated points: for $n \equiv 2 \pmod 6$, $M = \{n-2\}$ and $v_1 = 2(n-2)$ it is nonempty and has no component of positive length, being $\{1/8, 3/8, 5/8, 7/8\}$ at $n = 8$. The corresponding modifications $([n-1] \setminus \{n-2\}) \cup \{2(n-2)\}$ are tight, verified for $n = 8, 14, 20, 26, 32, 38, 44, 50$, and are members of the Goddyn–Wong families.

In the cases we examined with $k \ge 2$, the surviving component sits at a base point a/q with $q \in M \cup \{n\}$ and $\gcd(a, q) = 1$, displaced when $2q \le n-1$ by the presence of the speed $2q$. The number of such base points that can carry a component of the maximal length is up to $2k$, not $k+1$: each $b \in M$ coprime to n contributes its own tight point a/n with $a \equiv b^{-1} \pmod n$, as well as the points a/b. What is missing is an arithmetic criterion showing that k inserted speeds cannot block all of them at once.

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